

Reviews of Shipbuilding Economies

Peer Review of the United Kingdom Shipbuilding Industry



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Foreword

Over the past year, the OECD Shipbuilding Unit has been co-ordinating with the United Kingdom's National Shipbuilding Office on a programme of work to improve knowledge of and recommend policies to support the country's shipbuilding and ship repair industry. Research was conducted along four branches of inquiry: a data collection 'toolkit', the development of a framework to quantify skills imbalances, a comparative market and policy analysis, and identification of market failures. Various quantitative and qualitative approaches were used, including mapping available information, analysis of various key economic indicators, collection and analysis of additional data, in-depth literature research, and stakeholder engagement. This work has resulted in three reports: on data capabilities, on the quantification of skills imbalances, and on opportunities for and barriers to resilient growth.

Programme management and NSO-OECD co-ordination was undertaken by Matthew Jones and Leonie Finke on the NSO and OECD side, respectively, under the guidance of James Whitehouse Hebbourn and Laurent Daniel, respectively. Projects were led by Emilie Berger, Marina Daley, Yuki Inoue, and Leonie Finke, with contributions from Sunhye Lee, Juliana Gonzalez, Anna Lafforgue, Jiwon Choi, Hugo Vitrac, and Naveed Anjum.

The resulting reports were prepared by Leonie Finke, Yuki Inoue, Emilie Berger, Marina Daley, Anna Lafforgue, Jiwon Choi, and Hugo Vitrac, under the guidance and supervision of Laurent Daniel and Stephan Raes.

The authors would like to express their sincere thanks to Matthew Jones, James Whitehouse Hebbourn, Gavin Marshall, Rue McIlmoyle, James Hogg, and other colleagues at the UK National Shipbuilding Office (NSO) for their excellent collaboration, valuable input, and continued engagement throughout the development of these reports.

The authors would also like to thank the many stakeholders, whose time, trust, and perspectives have truly enlivened the narrative these reports have sought to uncover.

Finally, the authors thank the various OECD colleagues who were consulted throughout the research and writing process, enriching with diverse expertise the analysis and outcomes of these reports.

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Executive summary

Over the past year, the OECD Shipbuilding Unit has been co-ordinating with the United Kingdom's National Shipbuilding Office on a project to strengthen the evidence base and informing policy to support the country's shipbuilding and ship repair industry. This collaboration has resulted in three reports.

Informational capabilities in shipbuilding, abroad and in the UK (Reports I and II)

Data is a key issue which impacts knowledge, decision-making, and co-ordination at international, national, and nations' levels. Challenges and solutions exist along three domains: industry classification, data production and availability, and knowledge exchange. Addressing data issues will have far-reaching benefits for industry and government action (DSTI/SBC(2025)9).

A key challenge for the sector is the lack of harmonised and comprehensive information on skills and skills needs, which affects planning, workforce development, and the ability to match the right skills to the right place at the right time. Issues and potential solutions emerge across three areas: the definition of skills concepts, the availability and quality of data sources, and the methodologies used to assess skills needs. Addressing these gaps would provide significant benefits for industry and policymakers in the United Kingdom.

Opportunities and barriers to growth and resilience in UK shipbuilding (Report III)

In the non-defence space, the United Kingdom has faced a decline in its global market share and production capabilities. Globally, shipbuilding has grown in the twenty-first century; the key players have changed.

Resilience among peer shipbuilding economies (Norway, the Netherlands, and Italy) has tended to exist alongside specialisation into high-value segments, export orientation, and a positive reputation. The comparative analysis suggests the following market segments as potential opportunities for the UK.

Government rationales for intervention to realise these opportunities exist in regard to contextual market distortions, related industries, 'public needs' and public procurement, skilled labour, digital transformation, innovation and emerging technologies, and Net Zero.

For competitive specialisation to be successful, policy support for revitalisation of key points which influence production capabilities is needed: advanced facilities (digital transformation), skilled labour, a predictable and accommodating regulatory environment, enhanced innovation and commercialisation support, infrastructure development, supply chain security and a comprehensive approach towards cross-sectoral issues.

1 Introduction

This chapter sets out the purpose, scope, and methodological approach of the report, and summarises the key findings and conclusions across three core areas: i) informational capabilities relating to data availability on the shipbuilding sector; ii) comparative analysis of the UK and peer shipbuilding economies; and iii) opportunities and barriers to growth and resilience.

A. Introduction

This report presents key insights from the research undertaken on comparative informational capabilities, on improving data on skills needs in the United Kingdom's shipbuilding industry, and on identifying opportunities and challenges in the industry as well as possible industrial policy responses.

This analysis focuses on commercial shipbuilding in the UK and generally excludes shipbuilding for defence purposes. Although naval shipbuilding represents a much larger share of the UK's total production and has performed significantly better than the commercial segment, it falls outside the scope of the OECD's work and will only be referenced where it influences commercial activity. Because defence accounts for such a substantial portion of UK output, whether a data source includes or excludes defence production can materially affect the results. Throughout the paper, it will be clearly indicated when data points include (or exclude) defence shipbuilding. While commercial and defence shipbuilding share certain characteristics and resource requirements, such as reliance on highly skilled labour, defence activity is treated in this analysis as an external factor influencing conditions in the commercial sector.

The work on informational capabilities comprised two research projects: one on data capabilities (Report I), the other specifically on skills needs and their measurement (Report II). For the former, the research mapped definitional scope of the industry, economic indicators, and data availability both on a cross-country comparative level and nations level for the UK. For the latter, the research reviews theoretical and applied methodologies for defining occupational skills; for assessing skills supply, demand, imbalances, and mismatches, including through regional or cluster-based approaches; and for identifying future skills needs. Drawing on selected best practices, the project developed a measurement framework and applied it preliminarily to three occupations: welder, pipe fitter, and electrical control systems engineer. In both reports, engagement with stakeholders from the UK shipbuilding industry and related sectors strengthened the findings and ensured that the analysis reflected the domestic context. Key insights on comparative and the UK's informational capabilities are also covered in greater detail in the 'Data Collection Toolkit' Report (DSTI/SBC(2025)9), also submitted to the Shipbuilding Committee for the April 2025 meeting.

The work on comparative opportunities and challenges was conducted across two parallel workstreams, and results were merged for a comprehensive assessment and set of recommendations (Report III). Global market analysis was conducted to position the UK along various indicators of interest. Next, in-depth comparative market, industry, and policy analysis was conducted among peer shipbuilding economies of the UK: Norway, the Netherlands, Italy, and Spain. Opportunities and challenges were identified for key sub-sectors of interest to the UK – ferries, leisure vessels, and offshore wind vessels – as well as emerging technology-driven sub-sectors – green maritime, digital transformation, and autonomy. Market and government failures were explored throughout and were organised along key themes: contextual market distortions, public needs and procurement, skilled labour, innovation and technology, and Net Zero. Engagement with stakeholders and literature helped colour in questions left by data analysis (truncated by data availability challenges).

B. Informational capabilities

Significant differences exist across countries in the classification of what is considered part of the shipbuilding sector, and in the reporting of core industry indicators. While ship construction and repair are consistently covered, the inclusion of other upstream or related industries such as military shipbuilding, or marine equipment varies widely. Similarly, shipbuilding indicators – mapped across input, output, productivity and performance – are often inconsistently reported, and partially available. These differences hamper cross-country analysis and limit the effectiveness of policy making.

The report also highlights the fragmentation and inconsistencies in the collection of shipbuilding indicators in the UK. Although well-developed data collection structures exist, their effectiveness is undermined by classification errors, discrepancies in how economic activities are reported, and limited co-ordination between public and private actors at different levels of governance.

Data quality, availability and comparability are key issues impacting knowledge, decision-making, and co-ordination, on international, national, and nations' levels. The absence of systematic and comparable evidence significantly constrains the ability of government to anticipate structural risks, and to efficiently allocate public resources.

As the sector continues to innovate and decarbonise, skills needs are evolving rapidly, creating significant challenges for shipbuilding economies seeking to maintain competitiveness. A key obstacle is the absence of harmonised, comprehensive data on current and future skills requirements, which limits the sector's ability to secure the right skills, in the right place, at the right time.

Effective workforce planning in shipbuilding depends on a clear understanding of how skills needs are shaped across the sector. This requires consideration of several interrelated dimensions, including the availability of skills entering the labour market, the nature of employer demand, and how these evolve over time and across regions. Key aspects include skills supply through education and training pathways, skills demand reflected in employer needs and emerging technologies, the extent of skills imbalances and mismatches, regional clustering of capabilities, and the implications of technological, market and policy developments for future skills requirements.

C. The UK shipbuilding industry in a global context: lessons from peer economies and opportunities

Global shipbuilding has shifted towards East Asia in the twenty-first century. China now represents half of global production and an even greater share of the orderbook, followed by Korea and Japan. Productivity gains in these countries reflect an increase in the size of yards and vessels being built, and the deployment of advanced digital facilities. By contrast, the UK shipbuilding industry has contracted both in terms of production volumes and market share, and build times remain significantly higher than global averages.

Peer European shipbuilding economies demonstrate that resilience is possible through specialisation in high-value segments, export-oriented production and strong reputation. Facing similar challenges in competing against East Asian economies, some European shipbuilding countries have managed to secure certain segments of the global market.

Competitive specialisation opportunities exist for the UK, especially in three key segments – ferries, leisure and offshore wind vessels – as well as in emerging technology-driven subsectors, green and autonomous vessels. These areas build on demonstrated domestic strengths in design and innovation and align with growing global demand for low-emission and smart vessels, and with the expansion of international demand for offshore wind energy.

While specialisation has supported resilience in some peer economies, it has not always delivered long term growth. As rapid technological changes widen the gap between shipbuilding economies, specialisation in niche submarkets might not be enough to maintain the production capabilities and market shares. Investment in the modernisation of production infrastructures, and in emerging technologies will be essential for ensuring long term growth and resilience in the sector.

D. Realising opportunities

Realising these opportunities will require addressing market failures and distortions that constrain the UK's competitiveness, productivity and investment. The existence of these barriers –arising from unfair competition, spillovers from related domestic industries, the satisfaction of ‘public needs’ through public procurement, information and co-ordination in training and employment, digital transformation challenges, innovation and emerging technologies, and the transition to Net Zero – provide clear rationales for government interventions.

Policy interventions need to be mindful of contextual market distortions impacting the UK shipbuilding sector and the markets it operates in. Globally, varying level of subsidisation by governments and heightened concerns over supply chain security – especially in key inputs such as steel, carbon and glass fibre composites, and batteries – challenge the level playing field. Domestically, the interrelatedness of the shipbuilding sector with other advanced manufacturing industries creates both positive and negative spillovers that must be incorporated in the design of government interventions. While the shipbuilding industry in the UK benefits from a cluster effect with related industries that can pool skilled labour and generate knowledge spillovers, it suffers from indirect competition over shared inputs increasing their prices and undermining the sector's competitiveness, especially from the aerospace and the naval shipbuilding industry. These areas of competition for shared resources also represent ones where horizontal policies can positively impact multiple sectors.

Skilled labour shortages are one of the most acute challenges facing the shipbuilding industry. Market failures in the training and skill system create skill shortages and imbalances undermining firms' production capabilities and the sector's international competitiveness. At the company level, shipbuilders face difficulties attracting skilled workers, as they compete with related industries that often offer higher wages and benefit from a stronger reputation. High living costs in maritime clusters can further discourage prospective workers. Shipbuilders also incur additional costs when replacing experienced workers with younger recruits, as this can lead to short-term inefficiencies and requires substantial investment in training, particularly as new technologies continue to reshape production processes. These challenges are further exacerbated at the industry level by limited, fragmented information on current and future skills needs. This lack of visibility hampers efforts to modernise education and training systems and ultimately restricts the pool of skilled labour available to shipbuilders. Targeted policy interventions may be especially effective in addressing shortages for skill profiles shared across related industries, helping to ease competition for talent and strengthen the resilience of the broader maritime industrial workforce.

Digital transformation and innovation represent essential drivers of productivity growth, resilience and competitiveness, yet current evidence suggests uneven adoption across the UK shipbuilding sectors, and government failure in the design of policy support to innovation compounded by the little information on the “real” impact of these programmes. Several supply-side innovation programmes support R&D in the shipbuilding sector, but the innovation-to-commercialisation step remains a persistent bottleneck. Firms face fragmented information on available support, insufficient guidance beyond financial instruments, short funding cycles that constrain long-term planning, regulatory uncertainty, and limited co-ordination among investors. Strengthening policy design through closer co-ordination between government and industry, improving regulatory clarity and ensuring the availability of data on key indicators of the sector would help bridge the gap between innovation and commercial deployment.

Net zero policies are reshaping shipbuilding demand. Domestic and international commitments are creating new markets for low- and zero-emission vessels, alternative fuel technologies and reshaping the types of vessels being demanded. However, co-ordination failures between maritime actors, imperfect information on the regulatory pathway and unpriced environmental externalities slow down the pace of transition, creating clear rationales for targeted government interventions. New vessel types and markets

are emerging, but co-ordination and information failures, as well as negative externalities create rationales for government interventions.

Public procurement is a valuable tool to support innovation, and the net zero transition, with potential risks of adverse market effects. Used strategically, public procurement can serve as a key demand-pull instrument to support public goals and create early markets for emerging technologies, while addressing public needs. However, it can also come at the expense of productivity improvements, stimulate excess capacity and impose significant costs for companies participating in complex bidding procedures.

Strategies supporting shipbuilding embedded in broader regional and cross-sectoral development objectives may foster both growth and resilience through enhancing spillover effects and market dynamism. By acknowledging the embeddedness of shipbuilding companies within regional clusters, as well as their broader interrelatedness with other advanced manufacturing industries, policy makers can design interventions that may promote positive spillovers effects.

2 Informational capabilities

This chapter reviews the availability and quality of shipbuilding data from both domestic and international perspectives, highlighting key gaps that constrain effective policy design and workforce planning. It explores challenges related to industry classification, data production and knowledge exchange, and emphasises the role of reliable skills data in improving understanding of current and emerging workforce needs.

Data availability is a key issue which impacts knowledge, decision-making, and co-ordination, on international, national, and regional levels. Challenges exist along three domains: industry classification, data production and availability, and knowledge exchange. Addressing these data issues will have considerable benefits for industry and government action.

For the investigation of data capabilities across countries and within the UK, three domains were identified as key:

- **Industry classification** – ‘*what is the definition of the issue I have information on?*’ i.e. the definitional scope of the shipbuilding industry, how firms and economic activities are classified, and the approach towards emerging technologies.
- **Data production and availability** – ‘*what information do I have?*’ i.e. what information is produced in a systematic way by firms, government bodies, or other organisations, and published in a non-discriminatory fashion?
- **Knowledge exchange** – ‘*who do I share my information with?*’ i.e. what and how information is exchanged and organised through various systematic or ad-hoc mechanisms, across industry and government.

A. The cross-country lens

To better understand informational capabilities comparatively, formal industry classifications and accessible data for effective analysis were mapped across Australia, Canada, the People’s Republic of China (hereafter China), Spain, the European Union, the UK, Italy, Japan, South Korea, the Netherlands, Norway, and the United States.

Industry classification. Inconsistent industry definitions limit international comparability. A review of classification systems, as well as of the International Standard Industrial Classification (SIC) system, reveals significant differences in what is considered part of the shipbuilding sector. While ship construction and repair are consistently included, other elements, such as military shipbuilding, marine equipment, design, and port infrastructure, vary widely. These differences are often rooted in national industrial structures and historical context, but they hamper cross-country analysis and risk underrepresenting innovation and emerging technologies in policy assessments.

Data production and availability. Key industry indicators are inconsistently reported, making direct comparisons difficult. Shipbuilding performance was assessed across four indicator categories: input, output, productivity, and performance. Employment indicators vary considerably across countries, and output metrics are reported in different units even by major shipbuilding nations. Clear reporting on productivity is limited, and detailed performance indicators, including innovation and intellectual property measures, are only partially available. International organisations and commercial data providers fill some gaps, but more systematic reporting of indicators can support robust, comparable analysis.

B. The domestic lens

Strong infrastructure exists, but data collection is fragmented and inconsistently used. Despite having well-developed structures for data collection, the UK shipbuilding sector contains systemic areas for improvement. Firm-level classification errors, inconsistencies in the reporting of economic activities, and unco-ordinated efforts between government departments, regional clusters, and industry bodies result in gaps and duplication. While detailed data are collected at regional and sectoral levels, they are not always organised or accessible. Industry associations have become key points of knowledge exchange, but few have made structured, standardised data collection a priority.

Classification of firms and economic activities. Various solutions have been developed to account for challenges regarding firm categorisation, economic activity classification, and data availability. This includes the Data City's RSIC and RTIC AI-driven adaptations from the SIC system, the Scottish Industry Directories, and taxonomies developed by individual industry associations.

While these initiatives resolve challenges in sub-sector classification, especially in regard to emerging submarkets and technologies, a key issue is diversity which compromises comparability. When different systems of classifications or taxonomies are used, even empirically robust reports may produce analytical findings that differ significantly. In this way, solutions to improve information can also result in conflicting findings. That the choice in taxonomy is one which is often hidden deep within (unread) documentation of methodology can further obfuscate analysis across multiple sources.

Data production and availability. Diversity in data production capabilities and choice of indicators among firms and industry associations impact how well data sourced from across the industry can be harmonised or compared.

Data production capabilities by firms differ greatly and are generally correlated to the firms' degree of digitalisation: larger defence firms and those engaged in high-end technological capabilities typically have the infrastructure to collect continuous or frequent data points for a wide range of indicators related to monitoring vessel production and operation, factory settings and production processes, financing, and other broader social and economic factors. Conversely, SMEs, particularly those outside of defence or advanced technology submarkets, lack the infrastructure for that scale of data collection. As such, the data which could ultimately feed into various cross-firm collection mechanisms is itself created unevenly among different groups of firms. This limits the extent to which improvement of the data collection mechanism can comprehensively 'gather' data from firms.

Knowledge exchange and data sharing. Cluster and other industry associations have become a key node in the flow of knowledge between government and industry, as well as among individual firms. Even though clusters are well situated to gather information from members, few have made data a priority, and many rely on organic information rather than standardised systematic mechanisms.

The data collection capabilities of industry associations differ widely. For example, Maritime-UK Southwest has a database of circa 3,500 businesses through their newsletter. Their informational capabilities rely mostly on the organic use of information rather than more organised collection of data. Other cluster organisations rely on and pay for ad-hoc data collection and analysis through external services. Finally, in related industries, the ADS Group expands on official public data through firms' memberships submissions – firms directly send information on employment and apprenticeships upon membership registration, as well as turnover on an annual basis – and additional ad-hoc surveys.

One challenge highlighted by multiple stakeholders is competition for information between organisations which can result in survey fatigue. On the other hand, one of the characteristics of best practice examples is a combination of general data collection with verification on the ground.

C. Skills in focus

The lack of harmonised and comprehensive information on skills and skills needs is a significant barrier to ensuring the sector has access to the right skills, in the right place, at the right time. This challenge is particularly acute at a moment when interest in UK shipbuilding has intensified, creating a generational opportunity to align industrial ambition with workforce development and regional growth.

Current data systems are insufficient for effective workforce planning in shipbuilding. Existing metrics struggle to capture the sector's occupational complexity, regional variation and emerging skills requirements. Standard industrial and occupational classifications, such as SIC and SOC, tend to underestimate employment and obscure critical roles within shipbuilding and related activities. Publicly available datasets are often highly aggregated and uneven in coverage, with notable gaps across Scotland,

Wales and Northern Ireland, limiting the ability of policymakers and industry to anticipate and respond to localised skills pressures.

Addressing these limitations requires a more structured approach to assessing skills needs. An effective framework should analyse:

- **Skills supply:** Track vocational and higher education outputs, apprenticeships, and other qualification metrics.
- **Skills demand:** Monitor current vacancies and employer needs, while also anticipating future trends driven by technological advancements.
- **Skills imbalances and mismatches:** Compare available workforce skills against evolving job requirements to highlight both shortages and areas of over-qualification.
- **Clusters:** Examine geographical clusters (i.e. shipbuilding clusters and related industry clusters) to understand current and future skills availability.
- **Future skills demand:** Conduct skills foresighting and forecasting to understand how evolving technological advances, market trends and policy trends will affect future occupational demand.

The Shipbuilding Skills Needs Framework developed in Report II provides a structured response to these challenges, setting out methodologies that can be applied flexibly by organisations with differing analytical capacities. By combining quantitative indicators with occupational foresight and data gap analysis, the framework enables more granular, evidence-based workforce planning and clarifies both existing data coverage and remaining information gaps.

Sector collaboration and career visibility must improve. Improved data alone, however, is not sufficient. Stronger co-ordination across government, industry and education remains essential, including through existing mechanisms such as the Shipbuilding Skills Development Group. Greater clarity around career pathways, simplified qualification frameworks and more accessible training funding can help increase engagement, particularly among SMEs. At the same time, strengthening the visibility of shipbuilding as a technologically advanced, digital and decarbonising industry is critical to attracting younger and more diverse talent.

A resilient, future-ready shipbuilding workforce will not emerge by chance. It requires strategic planning, improved data, and targeted interventions that respond to both present and future needs. This report supports the National Shipbuilding Office and its partners in moving from reactive responses towards a more proactive, co-ordinated workforce strategy, helping to sustain the competitiveness, innovation and resilience of the UK shipbuilding sector over the coming decades.

D. Impact on analysis of opportunities and challenges

Significant gaps in data availability have impacted both the comparative analysis across peer competitors and the domestic analysis. This is a challenge covered further in the two companion reports on data and skills.

Because of the above identified challenges in data, there are certain limitations to the findings in the next sections. For the comparative analysis of opportunities, data availability has most strongly impacted findings for smaller vessels and for emerging subspecialisations. When combining data from two different providers – one with more comprehensive data on vessels above 100 GT, the other on vessels below 100 GT – because of differences in vessel and sub-sector categorisation, some judgment calls had to be made. Additionally, for aspects like ship design, autonomous vessels, and shipbuilding innovation, no official data sources provide sufficient information for quantitative trend or comparative analysis. For identification of market and government failures the analysis was primarily qualitative; the availability of empirical evidence to support presence or absence of challenges, not least to identify and quantify impacts of certain variables on the market, was relatively limited.

3

Global context and peer comparisons

This chapter situates the UK shipbuilding industry within global market trends and provides a comparative analysis with peer shipbuilding economies, Norway, the Netherlands, Italy and Spain. The chapter shows how growth and resilience among these countries have been linked to specialisation in high-value segments, export-oriented production and innovation. The chapter identifies key submarkets and emerging technologies that could offer similar opportunities for the UK.

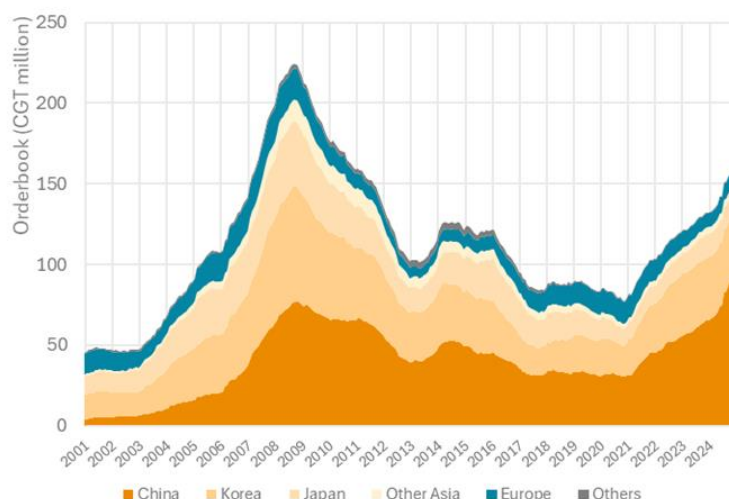
A. Global context

Globally, shipbuilding has grown significantly in the twenty-first century, but the key players have changed.

Production has increasingly moved towards Asia; China, Korea, and Japan are the largest shipbuilding economies along various indicators of output and performance. At the turn of the century, Korea, Japan, and Europe led the shipbuilding market in terms of orders by country and region. However, in approaching the 2010s, China's and Korea's orderbooks increased while other major players maintained a relatively steady number (Figure 3.1). Following the 2008 financial crisis, there was a general decline. Since 2021, there has been a steady increase in orders, with China capturing an even greater share (Figure 3.2).

In 2024, the global orderbook constituted 132.8 million CGT, of which China accounted for 66.4 million CGT (50.0%), Korea 38.9 million CGT (29.3%), Japan 14.4 million CGT (10.4%), and Europe 8.6 million CGT (6.5%).

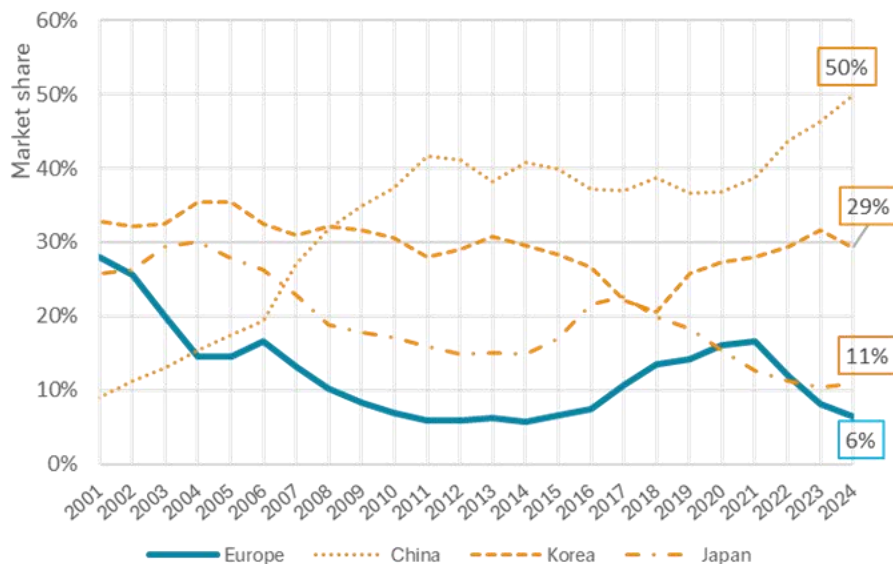
Figure 3.1. Monthly development of orderbook (2001-2024), millions CGT



Note: This graph includes sea-going vessel above 100 GT, excluding vessels built for the defence sector, not covered in Clarksons.

Source: OECD calculations based on World Fleet Register (Clarksons), and Shipping Intelligence Network, Clarksons research, <https://sin.clarksons.net/Home> [Access: February 2025].

Figure 3.2. Global market share of Europe, China, Korea, and Japan (2001-2024), based on CGT from orderbook



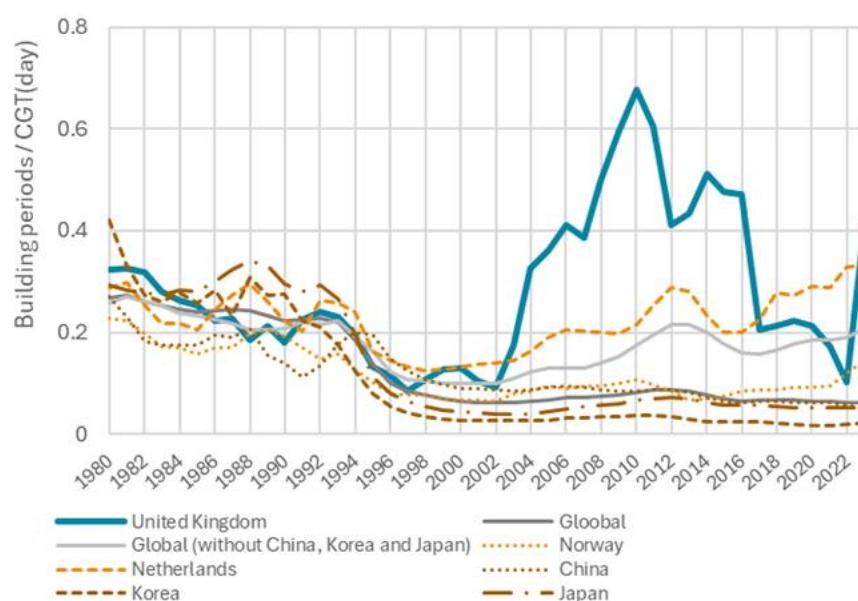
Note: This graph includes sea-going vessel above 100 GT.

Source: OECD calculations based on Shipping Intelligence Network (Clarksons).

In terms of global competitiveness, strategies adopted among top shipbuilding economies which could relate to productivity increases are order completions for larger ships, increased yard size, and introduction of advanced facilities.

In the non-defence space, the United Kingdom has faced a decline in its global market share and production capabilities. Whereas global shipbuilding volumes have increased significantly since 1990, the UK's shipbuilding volume has decreased in both absolute and relative terms. Similarly, while the average build time has decreased rapidly since 1990 (as build time/CGT), that of the UK has fluctuated and remained significantly higher than the global average (Figure 3.3).

Figure 3.3. Building periods per CGT (3-year average)



Note: This graph includes sea-going vessel above 100 GT.

Source: OECD calculations based on World Fleet Register (Clarksons).

In terms of global competitiveness, the UK has not kept pace with achieving cost reduction – productivity increases – associated with various potential factors: order completions for larger ships, yard size, and introduction of advanced facilities. Later sections also highlight the impact of an effective shortage in skilled, experienced labour.

Other European shipbuilding economies have secured certain segments of the global market despite facing similar challenges in competing against China, Japan, and Korea.

B. Country profiles

Country profiles: a snapshot

Across the board, resilience among peer shipbuilding economies tends to exist alongside specialisation into high-value segments and export orientation, as well as a positive reputation.

Norway has demonstrated resilience by swiftly transitioning to the production of cruise ships and ferries in response to the structural and global decline in demand for offshore oil vessels, which were previously their mainstay. However, it has not been sufficient to fully restore production to previous output levels. Moving forward, technological specialisation and a focus on emerging technologies, such as alternative fuels and automation, will be essential strategies for the industry to remain competitive and meet future market demands.

In the Netherlands, there has been a decline in vessel completions from 2000 to 2019. However, since 2020, the industry has shown signs of recovery, driven by steady output in specialised, high-value segments such as superyachts, dredgers, and custom cargo vessels. Dutch shipyards have increasingly focused on complex, technology-intensive vessels, leveraging their reputation for innovation and sustainability.

Italy has maintained a strong position throughout this period and has recovered well from the decline experienced between 2011 and 2015. This recovery can be attributed to the industry's ability to leverage its positive reputation and strong position in the cruise and superyacht manufacturing sectors, which has contributed to its current success. However, a potential weakness is that the industry's reliance on a few selected vessel types makes it susceptible to global structural changes in demand for these vessels. This highlights the need for some diversification to mitigate risks associated with fluctuating market demands.

Spain once had a very strong shipbuilding industry, but it has since declined somewhat. Unlike other peer countries, Spain did not adapt by shifting to vessel types with globally growing demand or by focusing on specific technologies or niche markets. As a result, the industry is now primarily producing fishing vessels for the domestic market, at least in the commercial vessel sector. This underscores the importance of actively focusing on and adapting vessel types to foster growth and recovery in the shipbuilding industry.

The UK experienced significant decline in (non-defence) production in the early 2000s, and production has since fluctuated at a relatively low level. One stand-out sub-sector is *leisure*, as *yachts* remain internationally competitive. However, broadly speaking this low level of production may both result from and reinforce declining production capabilities. Nonetheless, recent support measures point towards a strategy more in line with peer competitors: selectively developing specific sub-sectors, particularly through innovation support. One challenge the UK may face with this strategy is that it is a latecomer to these exact submarkets which other European countries have previously been proactively developing. Another is its negative reputation in select sub-sectors.

Norway

Norway has shown particular strengths in swiftly shifting between vessel types of interests to achieve targeted policy objectives – including Net Zero – and maintaining a mix of larger and smaller yards.

The industry

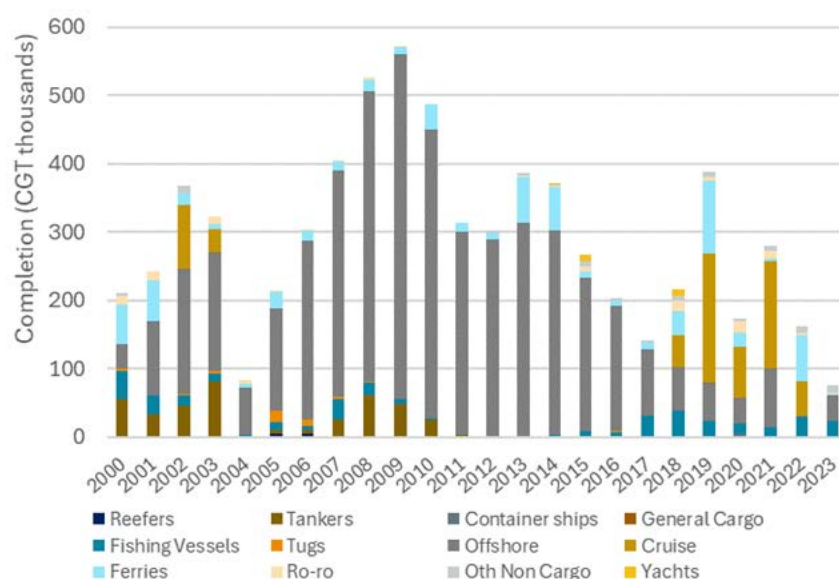
The Norwegian shipbuilding industry showed steady growth until 2009, with vessel completions measured in CGT peaking that year. Following the peak, diversity in production of vessel types decreased somewhat. Key vessel types have included offshore, cruise, and ferries.

A key submarket has been offshore vessels, associated with the strong performance of the global oil and gas market. Between 2006 and 2017, offshore service vessels accounted for 80% to 90% of Norway's annual vessel completions. However, after the sharp decline in global oil prices in 2014, the Norwegian offshore market contracted significantly. This downturn severely impacted Norway's shipbuilding industry, leading to a continuous decline in vessel completions until 2017 (Figure 3.4).

As offshore service vessel production declined, Norway re-diversified its production. This shift came from rapid increase in cruise and ferries completion. In particular, 2019 saw significant change, with cruise ships and ferries accounting for 82% of total completions, a remarkable growth from 5%-15% of total share before 2014. Notably, these completions in 2019 were ordered between 2016 and 2018, confirming that this shift occurred after – and likely as a response to, given the proactive policy measures put in place – the 2014 crash of oil prices. Although less prominent than cruise ships and ferries, fishing vessels also began to feature more in the portfolio of vessel completions from 2015, contributing to the diversification away from offshore vessels.

Despite such a shift, since 2019, the total number of vessel completions has gradually decreased again.

Figure 3.4. Vessel completions in Norway (2000-2023)



Note: This graph combines sea-going vessel from below 100 GT from Sea-web, with Clarksons World Fleet Register for sea-going vessel above 100 GT, excluding vessels built for the defence sector, not covered in Clarksons. The databases may not encompass all vessels worldwide, particularly those of a smaller size.

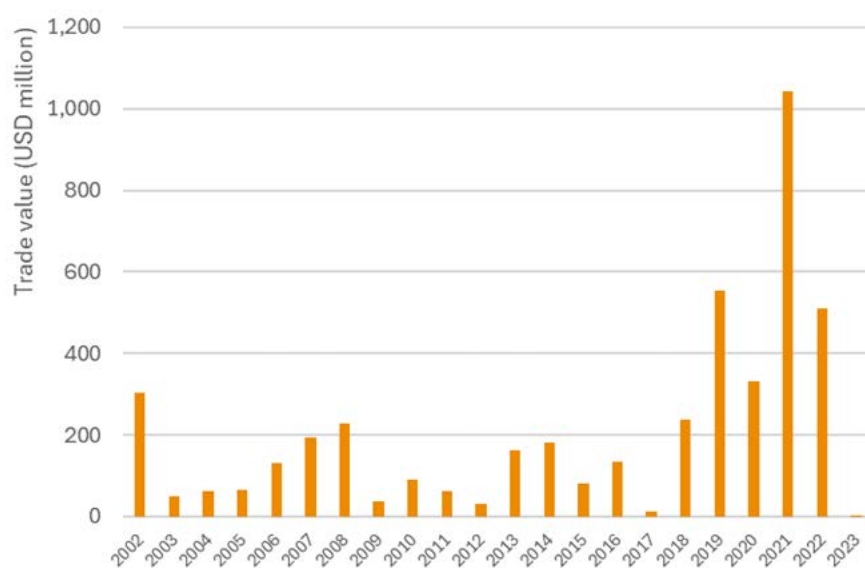
Source: OECD calculations based on World Fleet Register (Clarksons), and Sea-web, S&P Global, <https://www.spglobal.com/market-intelligence/en/industries/maritime> [Access: February 2025].

Exports and owner nationality

An export orientation emerged alongside the shift in production from offshore vessels to cruise and ferries.

Export data and vessel owner nationality provide important insights into the outward shift of Norway's shipbuilding industry in recent years. Between 2006 and 2016, when offshore oil dominated Norwegian shipbuilding, exports were minimal and more oriented towards, e.g. offshore platforms than other vessel types (Figure 3.5).

Figure 3.5. Export value of cruise, ferries, and offshore platforms produced in Norway (2002-2023)



Source: World Integrated Trade Solution(WITS), World Bank, <https://wits.worldbank.org/WITS/WITS/Restricted/Login.aspx#collapseSummary> [Access: February 2025]

This suggests that most vessels produced served the domestic offshore oil energy sector. This is supported by Figure 3.6, which shows the ratio of domestically owned offshore vessels which were also produced in Norway. The decline in this ratio correlates to the shift in regard to offshore oil around 2014.

Figure 3.6. Ratio of Norway-owned offshore vessels which were built domestically (3-year average)



Note: This graph includes sea-going vessel above 100 GT. This does not differentiate between structures and vessels for offshore oil versus those for offshore renewables.

Source: OECD calculations based on World Fleet Register (Clarksons).

After 2017, however, export values for cruise ships and ferries rose sharply, reflecting a growing reliance on these submarkets as outward-facing segments of the industry (Figure 3.5). This shift becomes even clearer when looking at nationality of ownership data. During the peak of offshore vessel production (2007 to 2016), approximately 50% to 80% of offshore vessels completed in Norway were delivered to domestic owners. In contrast, for cruise ships and passenger vessels produced between 2018 and 2023, —when these vessel types became the dominant products of Norwegian shipyards, —50% to 80% were built for foreign owners. This illustrates not only a clear shift in vessel types but also a broader transition towards export-oriented production (Figure 3.7).

Figure 3.7. Ratio of Norway-owned ferries which were built domestically (3-year average)



Note: This graph includes sea-going vessel above 100 GT. This does not differentiate between structures and vessels for offshore oil versus those for offshore renewables.

Source: OECD calculations based on World Fleet Register (Clarksons).

This structural shift in Norway's shipbuilding industry is closely tied to developments at Norway's largest shipbuilding group, STX Europe (formerly Aker until 2007) (European Commission, 2008^[1]). In 2013, STX Europe was acquired by the Italian shipbuilding giant Fincantieri (Fincantieri, 2023^[2]). Before the acquisition, STX Europe had not built any cruise ships or ferries in Norway. After Fincantieri's acquisition, however, the company – rebranded as VARD – emerged as the leading builder of cruise ships and passenger vessels in Norway (Clarkson Research, 2024^[3]). In fact, the majority of cruise ships completed in Norway in recent years have come from VARD's shipyards, underscoring the close link between foreign ownership, technological specialisation, and the growing export orientation of the Norwegian shipbuilding sector.¹

Policy perspective: support measures and other policy approaches

Norway provides proactive financial support, particularly in the form of R&D grants and loans.

The contribution of the Norwegian Export Credit Agency, Eksfin, is particularly significant. In addition to export credit support, Eksfin also manages the Domestic Ship Guarantee Scheme. This scheme was introduced in 2018 with an exposure limit of NOK 10 billion (GBP 921 million) and has been extended until 2026. Eksfin can cover medium- to long-term loans to purchase or carry out major conversions of ships

which are built wholly or partly at a shipyard in Norway and to be used in Norway. The maximum financial exposure at the end of 2023 amounts to NOK 1 276 million (GBP 97.2 million) (OECD, 2024^[4]).

There are also relevant sub-sector-specific support measures for green vessels, offshore wind, and autonomous vessels.

For ‘greening’ of the fleet, under its action plan for green shipping – which aims to reduce emissions from domestic shipping and fishing vessels by half by 2030 (Ministry of Climate and Environment, 2019^[5]) – Norway provided support through specific programmes such as risk loan schemes for short sea shipping and fishing vessels, and PILOT-E, a joint funding programme for innovation and commercialisation.

Norway has also rapidly strengthened its efforts in offshore wind development. In 2024, the Norwegian government proposed NOK 35 billion (GBP 2.55 billion) for a support scheme dedicated towards the first commercial floating offshore wind tender (offshoreWind.biz, 2024^[6]). This demand-side policy instrument is likely to stimulate further production of offshore wind vessels.

Finally, financial support in combination with other non-financial instruments have been applied in the autonomous vessel sub-sector. In 2016, Norway launched the world’s first test bed for autonomous ships (Kongsberg, 2016^[7]). The country has also established its own guidelines for autonomous vessels, in line with the policy objective of technological leadership in unmanned ships (Norwegian Maritime Authority, 2020^[8]). In 2022, the Yara Birkeland, widely regarded as the world’s first autonomous ship, entered commercial operation (YARA, 2025^[9]).

Key takeaways

The Norwegian shipbuilding industry has demonstrated resilience by swiftly transitioning to the production of cruise ships and ferries in response to the structural and global decline in demand for offshore oil vessels, which were previously their mainstay. However, it has not been sufficient to fully restore production to previous output levels. Moving forward, technological specialisation and a focus on emerging technologies, such as alternative fuels and automation, will be essential strategies for the industry to remain competitive and meet future market demands.

The Netherlands

The Netherlands has exhibited particular strengths in high-end technology and customisability in vessel design and production.

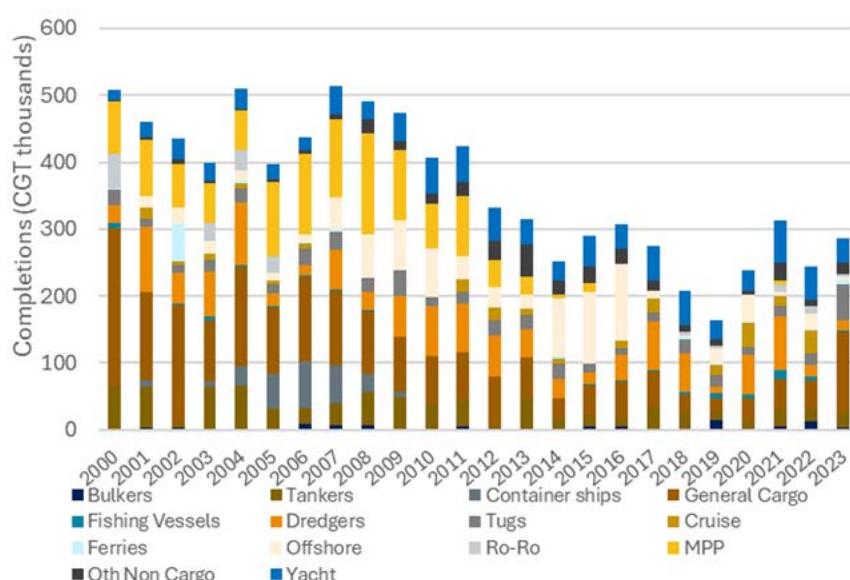
The industry

The Dutch shipbuilding industry experienced a gradual decline in vessel completions from 2000 to 2019, followed by a modest recovery in recent years. This decline can be partially attributed to decreased production of multipurpose vessels (MPPs) and oil offshore vessels, both of which had traditionally been important products for Dutch shipyards (Figure 3.8).

The drop in MPP orders stemmed from reduced demand from German and Dutch owners—the main historical buyers of Dutch-built MPPs. This decline reflects broader market shifts, including in response to the 2008–2009 financial crisis which triggered a slowdown in industrial trade and commodity flows, reducing the demand for MPPs. This trend was aggravated with falling charter rates, leaving shipowners with little incentive to order new MPPs (DNV, 2018^[10]). The collapse in oil prices after 2014 also lowered demand for MPPs used in offshore-related cargo transport, including the movement of drilling equipment and oilfield supplies.

Since 2020, Dutch shipbuilding has shown signs of recovery, driven by steady output in specialised, high-value segments such as superyachts, dredgers, and (custom) general cargo. Dutch shipyards have increasingly focused on complex, technology-intensive vessels—leveraging their reputation for producing high-quality vessels in a timely manner (OECD, 2020^[11]).

Figure 3.8. Vessel completions in the Netherlands (2000-2023), thousands CGT



Note: This graph combines sea-going vessel from below 100 GT from Sea-web, with World Fleet Register for sea-going vessel above 100 GT. The databases may not encompass all vessels worldwide, particularly those of a smaller size.
Source: OECD calculations based on World Fleet Register (Clarksons), and Sea-web (S&P).

Dutch dredgers are internationally competitive due to the country's leadership in innovative, sustainable technology. Companies such as Royal IHC, Van Oord, and Boskalis have introduced advanced solutions, including LNG-powered dredgers, deep dredging systems, and biofuels derived from waste streams, helping meet tightening environmental regulations. Dutch shipyards also benefit from public sector support, notably through the Ecoshape Consortium, which is backed by the Dutch Ministry of Infrastructure and Water Management and promotes the "Building with Nature" approach – the integration of natural processes into dredging projects. This combination of innovation and environmental leadership has helped Dutch dredgers maintain their appeal despite cost competition from Chinese yards (Van den Ende and Tarakci, 2018^[12]).

Similarly, the Netherlands has curated a reputation for high quality, timely delivery, and customisability in superyachts. Dutch shipyards benefit from a tightly connected ecosystem of suppliers, engineers, and designers, often referred to as "Yacht Valley," which supports the production of fully bespoke vessels (New Zealand Foreign Affairs & Trade, 2023^[13]).

By shifting towards niche markets and diversifying away from conventional MPVs, Dutch shipbuilders have adapted to changing global demand patterns. This strategic focus on high-value, custom vessels has likely supported the sector's gradual recovery.

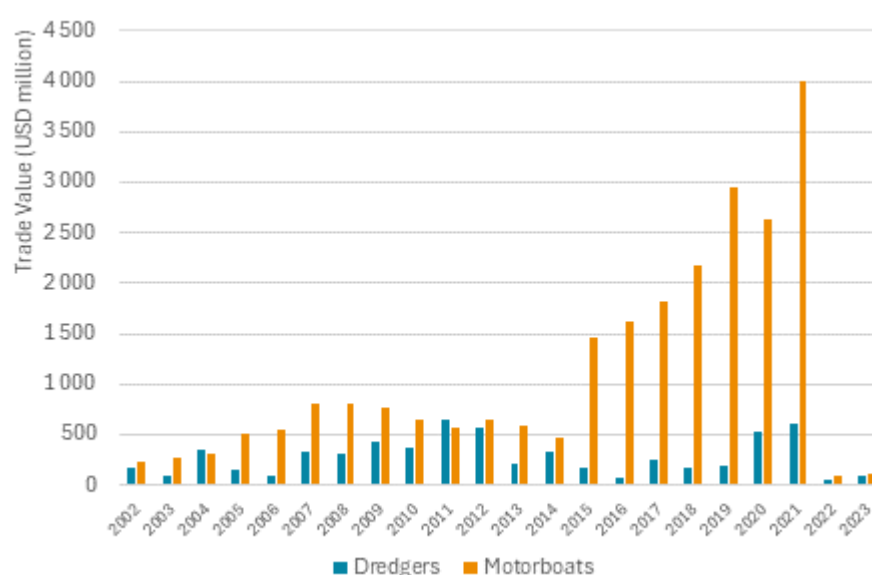
Exports and owner nationality

There is an increasing export orientation of the Dutch shipbuilding industry, particularly in segments where the Netherlands holds strong global competitiveness: dredgers and superyachts.

Historically, general cargo vessels built in the Netherlands primarily served domestic demand, but in recent years, more foreign owners have entered the order books. This is particularly true for dredgers, where two-thirds of vessels completed between 2000 and 2024 were for foreign owners (Clarkson Research, 2024^[3]).

Similarly, superyacht production is widely export-oriented. Recent trends show rising demand from younger, tech-oriented billionaires who want larger, technologically advanced yachts, including those equipped with sustainable propulsion and lightweight materials to improve efficiency. The combination of strong brand value, technological innovation, and environmental adaptation positions the Netherlands as a preferred source for both high-spec dredgers and luxury superyachts in the global market (Figure 3.9).

Figure 3.9. Export value of dredgers and motorboats



Note: Superyachts are included under the category of motorboats.

Source: World Integrated Trade Solution (World Bank).

Policy perspective: support measures and other policy approaches

In 2015, the government developed the Dutch Maritime Strategy 2015-2025 in co-operation with the maritime sector. It has aimed to address industry interconnections and societal trends. Key initiatives include establishing a Maritime Learning and Development Centre (MLDC), promoting exports, and supporting research, development, and innovation (RDI) (OECD, 2020^[11]).

Another mechanism for promoting co-ordination is SMASH!, a public-private partnership forum for smart shipping composed of companies, government actors, and knowledge institutions. The forum acts to promote multilevel governance and regional interconnectivity (SMASH!, 2021^[14]).

The Dutch government plays a supportive role in fostering innovation within the shipbuilding sector, particularly in areas where the industry holds technological leadership. One notable example is the Ecoshape Consortium, supported by the Ministry of Infrastructure and Water Management. This consortium focuses on advancing knowledge and technology for the 'Building with Nature approach,' which integrates natural processes into hydraulic engineering and dredging projects. Such public support for innovation helps Dutch shipyards maintain their competitive edge in environmentally sustainable dredging solutions, reinforcing the strong global reputation of the Dutch dredger industry (Van den Ende and Tarakci, 2018^[12]).

Relatedly, great emphasis has been placed on sustainability in shipbuilding and shipping. ‘Subsidy for Sustainable Shipbuilding’ is a scheme to encourage innovations for the construction or conversion of inland vessels, ocean-going vessels and offshore structures. The focus is on new, experimental technologies in the areas of sustainability, emission reduction, alternative fuels and noise reduction. The budget was EUR 2.3 million in 2023 (EGEN, 2021^[15]). Similarly, under the Subsidy Scheme for the Electrification of Inland Navigation Vessels, the Dutch government has made available a sum of EUR 15.1 million in 2024 and 2025 for the construction and conversion of inland navigation vessels capable of fully electric operation. The aim of this scheme is to grant subsidies for a ship to an electric propulsion line (EICB, n.d.^[16]).

Key takeaways

The Dutch shipbuilding industry experienced a decline in vessel completions from 2000 to 2019. However, since 2020, the industry has shown signs of recovery, driven by steady output in specialised, high-value segments such as superyachts, dredgers, and custom cargo vessels. Dutch shipyards have increasingly focused on complex, technology-intensive vessels, leveraging their reputation for innovation and sustainability.

Additionally, the Dutch government has played a supportive role in fostering innovation and sustainability within the sector, through schemes like the Subsidy for Sustainable Shipbuilding and the Subsidy Scheme for the Electrification of Inland Navigation Vessels. This demonstrates how focusing on niche markets, leveraging the country's innovation capacity, and receiving targeted government support can yield significant results.

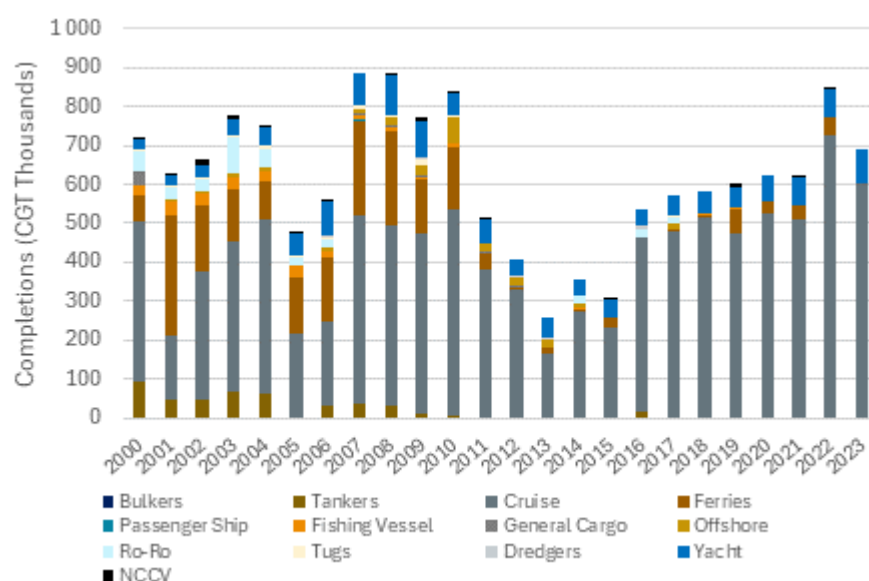
Italy

In its progressive position in the cruise submarket, in addition to export-oriented leisure, Italy is a good example for targeted specialisation as well as greening of vessels.

The industry

Generally, the Italian shipbuilding industry has maintained its strong position through a strong domestic supply chain that involves a network of specialised local enterprises. This has not only supported technological advancements but also contributed to employment and economic stability (OECD, 2024^[17]).

Production peaked in the late 2000s, then declined sharply (Figure 3.10). The industry began recovering in 2016, though the number of vessels completed did not increase significantly, suggesting a shift toward the construction of larger, more complex ships (Clarkson Research, 2024^[3]). A brief decline occurred during the COVID-19 pandemic, particularly in the cruise ship sector, but Italian shipbuilders successfully mitigated order cancellations through negotiated postponements (OECD, 2024^[17]).

Figure 3.10. Vessel completions in Italy (2000-2023), thousands CGT

Note: This graph combines sea-going vessel from below 100 GT from Sea-web, with World Fleet Register for sea-going vessel above 100 GT. The databases may not encompass all vessels worldwide, particularly those of a smaller size.
Source: OECD calculations based on World Fleet Register (Clarksons), and Sea-web (S&P).

Italy is a global leader in cruise ship construction, and despite the downturn between 2013 and 2015, cruise ship output has since rebounded, reaffirming its strength within the Italian shipbuilding sector. In addition to cruise ships, Italy also produces ferries, though this segment has become increasingly marginal. Since 2011, Fincantieri, the country's largest shipbuilder, has built only one ferry in Italy. Smaller shipyards, such as Cantiere Navale Visentini, have continued ferry production on a limited scale.

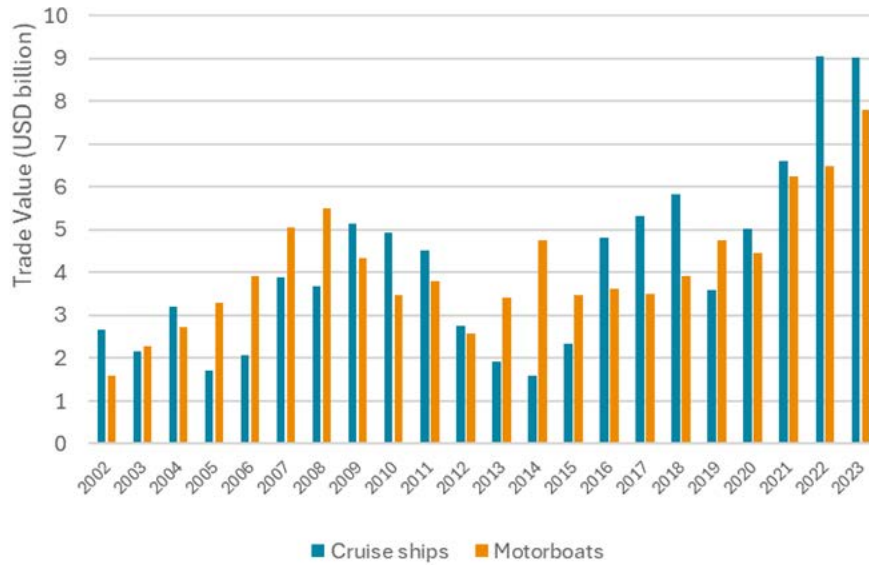
A defining feature of the Italian shipbuilding industry is its increasing specialisation. The industry has proactively focused on high-value, complex ships – namely cruise ships and yachts. This has reinforced its global competitiveness, particularly in the luxury and passenger vessel markets.

Exports and owner nationality

Italy's shipbuilding industry, particularly its export performance, reflects the country's global leadership in cruise ship construction. Since 2016, Italian cruise ship exports have grown steadily, closely following the recovery in cruise ship completions after the 2013-2015 downturn. Cruise ships produced in Italy are overwhelmingly destined for foreign operators, with more than 90% of cruise vessels completed between 2000 and 2024 delivered to international owners. This highlights the strong export orientation of Italy's cruise ship sector.

Yachts – classified as motorboats in the trade data below (Figure 3.11) – also comprise a major export product of Italy. 90% of the domestically produced boats (including sailboats) are exported, with 40% of export within the EU and 60% outside of the EU, namely North America. Italy is a top exporter of boats with inboard motors, with 26% of global market share (Confindustria Nautica Statistics Department, 2024^[18]).

Figure 3.11. Export value of cruise ships and motorboats built in Italy (2002-2023), billions USD



Source: World Integrated Trade Solution (World Bank)

However, the export orientation is much less pronounced for ferries. As seen in Figure 3.12 and Figure 3.13, before the financial crisis ferries built in Italy were primarily for domestic demand. This difference in market orientation likely influenced strategic decisions within the industry, particularly for Fincantieri. As Italy's domestic economy faced challenges around 2011, including weaker public and private sector purchasing power, the demand for domestically built ferries may have declined. In response, Fincantieri shifted its focus toward the more lucrative and stable global cruise market, where international demand remained strong. This shift underscores how Italy's shipbuilding industry—much like those in other European countries—has increasingly prioritised high-value, export-oriented production to maintain its competitiveness and economic resilience.

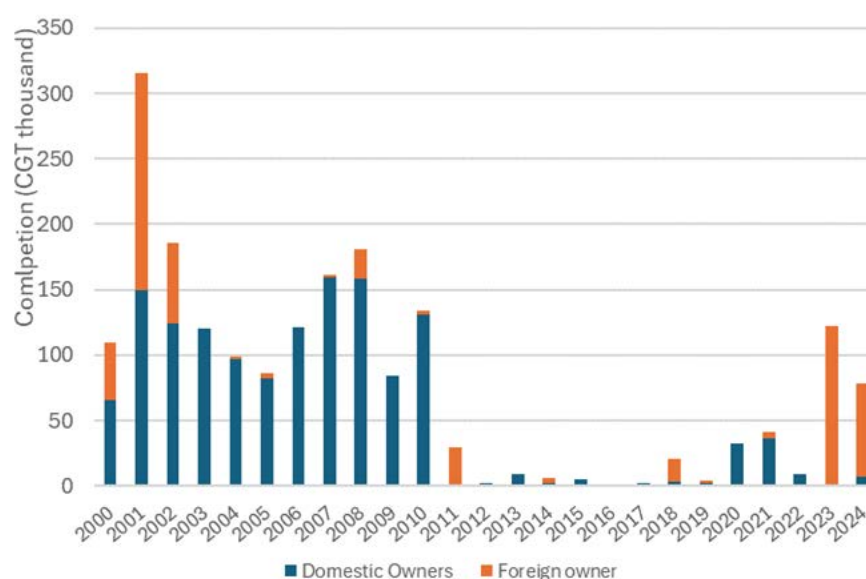
Figure 3.12. Ratio of Italian-owned ferries which were built domestically (2001-2023), %



Note: This graph includes sea-going vessel above 100 GT.

Source: OECD calculations based on World Fleet Register (Clarksons).

Figure 3.13. Development of domestic owner vs. foreign owner of ferries built in Italy



Note: This graph includes sea-going vessel above 100 GT.

Source: OECD calculations based on World Fleet Register (Clarksons).

Policy perspective: support measures and other policy approaches

In the twentieth century, during the 1980s and 1990s, the Italian government promulgated a number of specific laws to intervene in the shipbuilding sector which were mainly oriented towards the restructuring of shipyards. Following the establishment of the European Union and the adoption of the euro, Italian

legislation was brought in line with European rules, notably state aid rules, which prevent market-distorting policy interventions (OECD, 2024^[17]).

The Italian government has been making an effort to improve transparency on government support. In 2017, Italy published legislation which requires all companies receiving, “grants, subsidies, advantages, contributions or aid, in cash or in-kind, which do not have any reciprocal, remunerative, or compensatory nature” to publish the amounts they obtained from Italian authorities in the preceding year in the supplementary notes of their financial statements and any other consolidated financial statements (OECD, 2024^[17]).

Next, as a measure to contribute to the achievement of the objectives of the European Green Deal and Fit for 55 package, the Italian government launched a scheme to give direct grants to shipping companies registered in Italy to acquire clean and zero-emission vessels, as well as to retrofit the vessels. The budget of the scheme amounts to EUR 500 million (GBP 420 million) for the period 2021-2026 (European Commission, 2022^[19]).

Finally, in support of its strong marine tourism industry, local government often support this industry through packaged support measures for multiple co-located sectors. For example, the ‘Marina Arcipelago Toscano di Piombino’ project aims to transform an abandoned area into an eco-friendly nautical reception centre, co-financed by the Tuscany Region and the Regional Programme Fund. It will provide spaces for boat berths, dry ports, and facilities for marine tourism, water sports, local leisure, and shipyards (Marina World, n.d.^[20]).

Key takeaways

The Italian shipbuilding industry has maintained a strong position throughout the observation period and has recovered well from the decline experienced between 2011 and 2015. This recovery can be attributed to the industry’s ability to leverage its renowned name and strong position in the cruise and superyacht manufacturing sectors, which has contributed to its current success. However, a potential weakness is that the industry’s reliance on a few selected vessel types makes it susceptible to global structural changes in demand for these vessels. This highlights the need for *some* diversification to mitigate risks associated with fluctuating market demands.

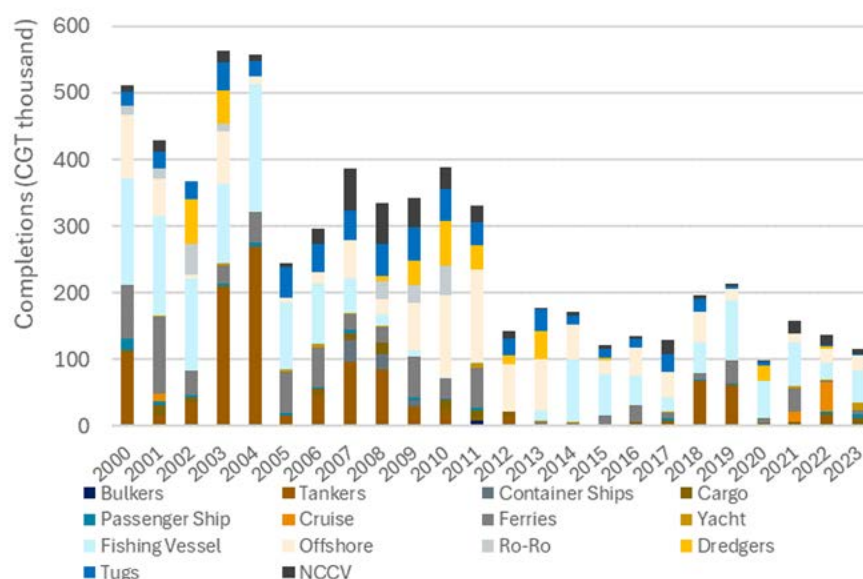
Spain

Though it still has some leading firms and solid production of fishing vessels, overall Spain has experienced a decline in market volume.

The industry

The Spanish shipbuilding industry has experienced a continuous decline since 2000, with a brief period of recovery between 2007 and 2011 (Figure 3.14). During this short rebound, offshore vessel production increased slightly, helping to stabilise the sector temporarily. However, the overall trend of contraction implies a decreasing competitiveness.

The most visible sign of this loss of competitiveness is the sharp decline in tanker production after 2004. Tankers once formed a substantial part of Spain’s shipbuilding portfolio, and their decline significantly reduced overall vessel completions. After 2014, when the global offshore market also contracted, the Spanish industry became increasingly reliant on fishing vessels, which have remained the primary type of merchant vessel produced.

Figure 3.14. Vessel completion in Spain (2000-2023), thousands CGT

Note: This graph combines sea-going vessel from below 100 GT from Sea-web, with World Fleet Register for sea-going vessel above 100 GT. The databases may not encompass all vessels worldwide, particularly those of a smaller size.
Source: OECD calculations based on World Fleet Register (Clarksons), and Sea-web (S&P).

It is important to note that these figures exclude naval vessels, as the data focuses only on merchant and commercial shipbuilding. Navantia, Spain's largest shipbuilder, is a state-owned enterprise primarily focused on naval shipbuilding and defence projects, meaning its core business falls outside the scope of this analysis.

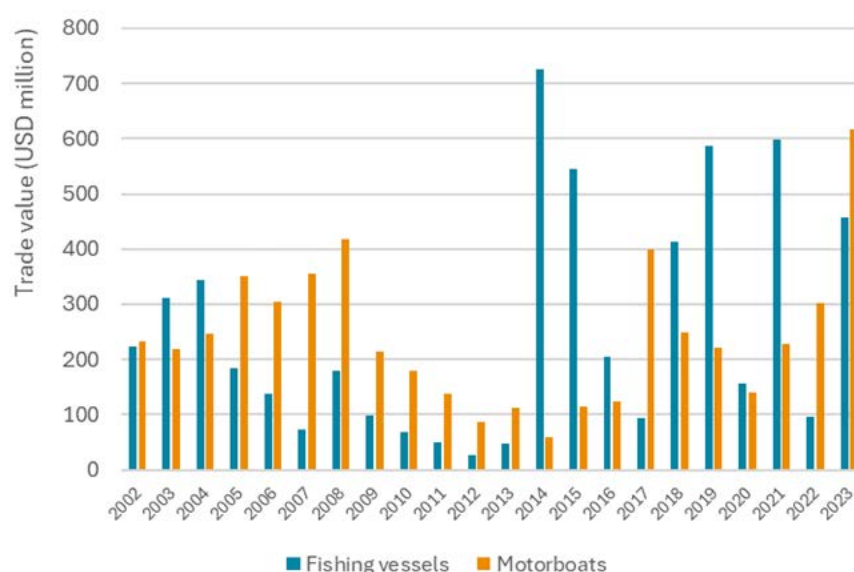
Exports and owner nationality

The Spanish shipbuilding industry has historically been outward-oriented, with exports playing a central role. Between 2000 and 2023, the weighted average share of vessels built for Spanish owners was approximately 27%, meaning around 73% of vessels completed in Spain were for foreign operators. This indicates that foreign demand has traditionally been essential to the industry's survival.

However, in recent years, the product mix of Spanish shipbuilding has shifted heavily toward fishing vessels and fish farm support vessels, and this shift has been associated with a change in customer base. In these segments, domestic demand dominates approximately 90% of fishing vessels and 67% of fish farm support vessels completed in Spain were delivered to Spanish owners. However, interestingly, export trade value data reveals that fishing vessels are still the most prominent product Spain exports in the shipbuilding sector (Figure 3.15). This highlights that Spain's fishing vessel sector does retain some export relevance, but the vessels exported are generally smaller, lower-value ships compared to the larger and more complex vessels once produced for foreign markets. In fact, compared to Italy and other peer countries, the scale of fishing vessel exports is still small.

Motorboats are also a main export product from Spanish shipbuilding industry, with trade value rapidly increasing in recent years. In 2023, Spain ranked 9th in global inboard boats export with 2% of global market share (Confindustria Nautica Statistics Department, 2024^[18]).

Figure 3.15. Export value of fishing vessels and motorboats built in Spain (2002-2023), thousands CGT



Source: World Integrated Trade Solution (World Bank)

Policy perspective: support measures and other policy approaches

Despite an overall decline, the Spanish government has still included support for shipbuilding in broader policy strategies.

In 2021, Spain launched Proyecto Estratégico para la Recuperación y Transformación Económica (Strategic projects for economic recovery and transformation, PERTE). PERTE is a strategic mission with a great capacity to boost economic growth, employment and the competitiveness of the Spanish economy, with a high level of public-private collaboration and application to the different administrations (Spanish Government, 2021^[21]).

PERTE contains 12 strategic projects, including on the shipbuilding industry. This project aims to diversify the naval sector towards new products, promote digitalisation, and improve environmental sustainability and skilled labour.

A total investment of EUR 1 460 million (GBP 1 255 million) is planned, with a public contribution of EUR 310 million (GBP 267 million) and an estimated EUR 1 150 million (GBP 989 million) of private investment. An investment aiming line will be launched for the modernisation and diversification of the sector and another dedicated to naval application technologies (Spanish Government, 2022^[22]).

Key takeaways

Spain once had a very strong shipbuilding industry, but it has since declined somewhat. Unlike other peer countries, Spain did not adapt by shifting to vessel types with globally growing demand or by focusing on specific technologies or niche markets. As a result, the industry is now primarily producing fishing vessels for the domestic market, at least in the commercial vessel sector. This underscores the importance of actively focusing on and adapting vessel types to foster growth and recovery in the shipbuilding industry. Additionally, it highlights the potential role of government guidance in steering the market towards more promising directions.

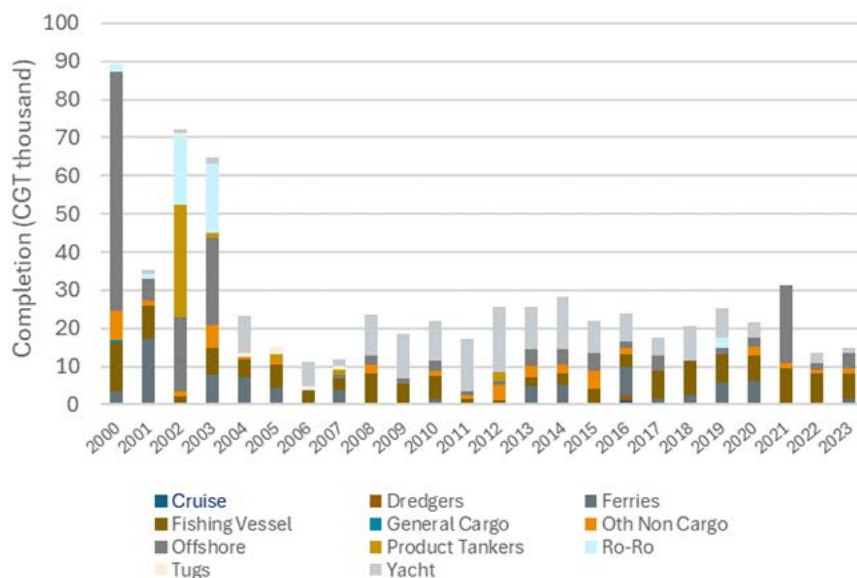
The UK

Outside of defence shipbuilding, the industry is comprised of a combination of aging shipyards, select successful start-ups, and firms that have become somewhat self-conscious in respect to their advanced manufacturing neighbours and foreign peers. Yet, there are key submarkets and existing strengths for realising current and future opportunities.

The industry

The UK shipbuilding industry has undergone significant transformation since the early 2000s, with a clear reduction in commercial vessel completions after 2003 (Figure 3.16). This trend reflects broader shifts in the global shipbuilding landscape, where competition from cost-effective East Asian yards has intensified – a challenge also seen across the other European countries profiled above.

Figure 3.16. Vessel completions in the UK (2000-2023), thousands CGT



Note: This graph combines sea-going vessel from below 100 GT from Sea-web, with World Fleet Register for sea-going vessel above 100 GT. The databases may not encompass all vessels worldwide, particularly those of a smaller size.

Source: OECD calculations based on World Fleet Register (Clarksons), and Sea-web (S&P).

Output has largely consisted of offshore, leisure, ferry, and fishing vessels. Notably, offshore structures and vessels comprised a key portion of completions in the early 2000s before sharply declining and only reemerging in recent years. Yachts, in particular, have maintained steady output, reflecting the UK's strong reputation for craftsmanship and customisation in the luxury leisure sub-sector.

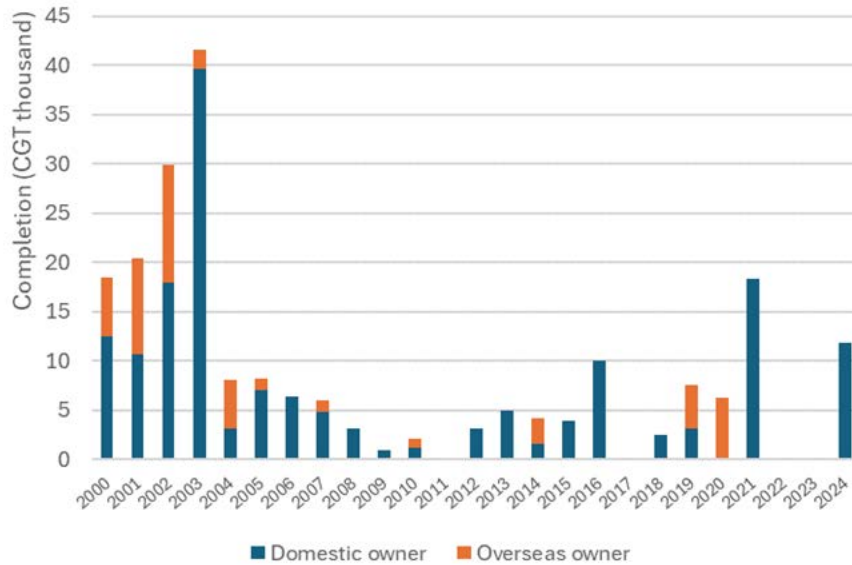
It may be relevant to note that Figure 3.16 has weak representation for the smaller vessels, especially those below 100 GT. Within the submarkets highlighted as relative strengths, production of vessels below 100 GT may be more common and would therefore not be visible here. The export data below, which includes smaller vessels, suggests that certain strengths remain.

Also, as mentioned previously, the UK's naval sector, championed by firms such as BAE Systems and Babcock, remains relatively stable through a combination of defence procurement and exports.

Exports and owner nationality

Generally, the industry has been relatively inward facing. Approximately 75% of the vessels completed between 2000 and 2024 (measured in CGT) were built for domestic owners. This indicates that much of the industry's output serves the needs of the domestic market. Nonetheless, as shown further below in Section 5.4, even in key submarkets the UK is unable to consistently capture significant portions of the total domestic demand (Figure 3.17).

Figure 3.17. Domestic vs. Overseas owner of vessels built in the UK (CGT)



Note: This graph includes sea-going vessel above 100 GT.

Source: OECD calculations based on World Fleet Register (Clarksons).

Simultaneously trade value data—which also captures smaller vessels not always visible in completion databases—shows that the UK maintains strong global competitiveness in certain segments. The UK ranks 5th in the global inboard motorboat market, holding a 5.6% share. In the sailing boat and yacht segment, the UK performs particularly well, ranking first in the small vessel category (up to 7.5 meters in length) and fourth in the superyacht market (orderbook) (Confindustria Nautica Statistics Department, 2024^[18]). The detail of the export of leisure boats is discussed in the key submarkets part.

Policy perspective: support measures and other policy approaches

The UK government has supported maritime sectors via at least an estimated GBP 257 million (see Table 3.1).

Table 3.1. Key ongoing support measures in the UK (2025)

Name of measures	Agency	Outline of the measures
UK Shipping Office for Reducing Emissions (UK SHORE) (Department for Transport. et al., 2022 ^[23])	UK SHORE (Department for Transport) and Innovate UK	UK SHORE was established in 2022 with GBP 206 million new funding to accelerate research into and development of clean maritime technologies and create skilled jobs across the country.
Clean Maritime Demonstration Competition (CMD C) (Department for Transport and Innovate UK, 2021 ^[24])	Department for Transport and Innovate UK	CDMC is a multi-year competition for research and development match-funding to support the design and development of clean maritime solutions towards commercialisation, launched in 2022.
Smart Shipping Acceleration Fund (Innovate UK and Department for Transport, 2024 ^[25])	Department for Transport and Innovate UK	The aim of this competition is to fund, up to GBP 8 million, detailed feasibility studies for innovative technology demonstrations of scalable smart shipping technologies that reduce greenhouse gas emissions and improve air quality.
Marine Supply Chain Accelerator (Crown Estate, 2024 ^[26])	Crowne Estate	Crown Estate have established a GBP 50 million Supply Chain Accelerator fund with the explicit objective of helping catalyse the UK supply chain capacity and capability for offshore wind.
International Green Corridors Fund: UK – IE Feasibility Studies (Innovate UK. et al., 2024 ^[27])	Department for Transport and the Marine Institute, Ireland	The aim of this competition is to fund, up to GBP 860 000, innovative feasibility studies into routes that stimulate early adoption of promising long-term solutions to reach zero emissions in shipping.

Note: The CMD C and Smart Shipping Acceleration Fund, amongst others, are situated under the UK SHORE programme.

Source: (Department for Transport. et al., 2022^[23]; Innovate UK and Department for Transport, 2024^[25]).

A thorough examination of these policies in their totality reveals that the UK government is implementing R&D measures to foster innovation within the shipbuilding industry. Other peer countries are also following a similar trend, i.e. R&D supports. However, to compete with these countries, it may be necessary for the UK to provide more in-depth support. For instance, the implementation of policies that are more closely aligned with social implementation, such as the “Marine Supply Chain Accelerator”, may be pivotal in this regard.

Box 3.1. Key policy instrument: the Clean Maritime Demonstration Competition (CMDC)

The CMDC is a key supply-side stimulant to innovation for green maritime technology

55 projects assessed and identified by Innovate UK received GBP 23 million in the first round of the CMDC to support green innovation notably in the shipbuilding sector. The last call CMDC6 was launched in January 2025 allocating GBP 30 million, bringing the total funding of the CMDC to GBP 159 million. The funding of this programme comes from the UK Shipping Office for Reducing Emissions (UK SHORE) programme (Department for Transport and Innovate UK, 2021^[24]).

There are two strands under the CMDC:

1. **Feasibility studies for projects** costing between GBP 25 000 to 1 million. Projects must review the technical and economic feasibility of technological solutions to reducing carbon emissions and develop a clear plan for future demonstration after the feasibility study has been completed.
2. **Innovation projects** costing between GBP 100 000 to 4 million. Projects must design, develop and then test a novel clean or smart maritime technology aiming to reduced carbon emissions, either from vessels, or from onshore infrastructures (Department for Transport and Innovate UK, 2021^[24]).

Successes: These funds have notably been used to support start-ups which are now in various phases of commercialisation. This includes, e.g. Artemis Technologies and DRIFT.

Remaining challenges: one issue raised by stakeholders, even those who have overcome it, is that the innovation to commercialisation step remains distinctly formidable, even with the CMDC, and that the short funding periods of this programme can be limiting. This may be where reliance on supply chains exists or development may have multiple phases.

Opportunities in key submarkets'

Key submarkets were identified by where there is (1) an interest in maintaining existing strengths, and/or (2) high growth potential due to global or domestic incoming geopolitical or economic shifts. The analysis identified ferries, leisure boats and offshore support vessels as key submarkets.

Key submarkets: a snapshot

Ferries. There is significant demand-side and supply-side opportunity to combine replacement of the aging public ferry fleet with even greater support for transforming innovation capabilities into production capabilities in the green vessel sub-sector.

- The public ferry fleet is aging and needs to be replaced.
- The domestic sub-sector for non-green ferries is characterised by lagging production capabilities and lingering effects of previous failures. Global competitiveness is negligible and public procurement bids often go abroad because of a face-value costing differences.
- The UK has strengths in green innovation but the innovation to commercialisation step is a challenge. Still, there is preliminary momentum in domestic green ferry or other vessel production. Some select start-ups focused on high-tech green vessels, including a ferry, have successfully commercialised through a combination of public and private investment. Delivery of initial builds in the coming year may set the tone for other entrants.
- Norway's proactive transition to a green ferry fleet can serve as a successful example.

Leisure vessels. The supply-side opportunity is that this sub-sector has a relatively strong position globally which could be further developed and used for resilience in the sector.

- The demand for leisure vessels (as a luxury good) may be treated as relatively stable under current circumstances.
- The leisure sub-sector highlights the benefits of co-location with related sectors and sub-sectors in industry clusters, within the UK but especially within Italy and the Netherlands, who both outperform the UK. Policy focused on place-based and cluster-wide impacts may be a factor of success for these key competitors.
- Other factors of success may be technological capabilities, skilled labour, and availability of key input materials.

Offshore wind vessels. There is a significant demand-side opportunity in the nascent offshore wind vessel submarket, domestically and abroad. Existing strengths in design and innovation would need to be transformed into robust production capabilities.

- The market for offshore wind development is growing, implying relatively certain demand in offshore wind vessels.
- While the UK has a relatively strong position in terms of ownership of these vessels, production is disproportionately smaller – absent for SOVs and CSOVs.
- An appropriate peer country to compare is Norway, which has significant positions in both ownership and production. While both Norway and the UK have design strengths, Norway far outpaces the UK in terms of its production capabilities for these vessels.
- Design and innovation strengths exist in the form of globally competitive firms and existing support measures, and some high-tech capabilities exist within the leisure sub-sector as well as through recent start-ups originating out of support schemes.

Additionally, emerging submarkets and innovation were investigated.

Innovation and emerging technology: a snapshot

The United Kingdom may have strengths in both the design of made-to-order products as well as in development of new products or technology in this field. Empirical evidence on design as a service sub-sector is highly limited but the success of various firms providing this service warrants further investigation. Support for innovation, particularly in green maritime technology, is relatively robust.

Two emerging sub-sectors offer opportunities in UK shipbuilding. Innovation capabilities in green technologies and autonomy have proven to be relatively competitive globally, and there are success stories of select start-ups, but the innovation to commercialisation step remains a challenge.

- **Green vessels.** Shipbuilding and shipping are moving towards maritime decarbonisation. Greening vessels includes the development and adoption of various technological applications which could be applied across existing sub-sectors (vessel types). The relatively nascent market offers opportunities for first mover advantages. Though domestic production of green vessels is lagging, the UK has existing strengths in the innovation of new green maritime technology. Because the transition is one which may occur across sub-sectors, strengths in green vessel production may help reverse the decline in production capabilities.
- **Autonomous vessels.** The autonomous vessel submarket is still nascent but can be expected to grow rapidly in the near future. The United Kingdom has a strong position in innovation and investment. Through a mission-oriented strategy, Korea is a prime competitor in this sub-sector, with well-co-ordinated investment, regulation, and dependable policy outlooks.

Key takeaways

Since a significant decline in the early 2000s, production of (non-defence) vessels in the UK has fluctuated at a relatively low level. One stand-out sub-sector is leisure, as yachts remain internationally competitive. However, broadly speaking this low level of production may both result from and reinforce declining production capabilities. Nonetheless, recent support measures point towards a strategy more in line with peer competitors: selectively developing specific sub-sectors, particularly through innovation support. One challenge the UK may face with this strategy is that it is a latecomer to these exact submarkets which other European countries have previously been proactively developing. Another is its negative reputation in select sub-sectors.

C. Conclusions

Global shipbuilding production has grown and shifted in nature and location.

Across the board, resilience among peer shipbuilding economies tends to exist alongside specialisation into high-value segments and export orientation, as well as a positive reputation.

- The Norwegian industry has focused on cruise ships and ferries;
- the Dutch industry recovered through innovation and high-tech high-quality vessels; and
- the Italian industry strengthened the reliance on its strong position in cruise and superyacht manufacturing.

In contrast, Spain's industry has struggled due to a lack of adaptation.

This suggests that for the UK, concentrating on established niche markets, along with emerging vessel types and technologies, may be essential.

However, two cautionary considerations exist. First, while this strategy appears to be associated with resilience, reflected in these countries maintaining a core share of key sub-markets, it has not been consistently linked to sustained long-term growth. Second, the industry may be entering a period of rapid technological change. Strategies that were sufficient in the past may prove inadequate as technological gaps between countries continue to widen.

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Note

¹ The value added and output used in STAN and UNIDO concerns the economic activity conducted within the border, regardless of the nationality of the economic entity. The aggregate of value added of each industry forms GDP. Therefore, the nationality of the company does not affect the GDP as long as the economic activity is conducted within the border.

4 Opportunities and barriers

This chapter examines opportunities for government intervention to support the UK shipbuilding industry, using market failures as a guiding framework. The chapter discusses rationales for government interventions to realise these opportunities and identifies priority areas where policy support can be most effective in enabling competitive specialisation.

Whereas the previous chapter assessed opportunities for UK shipbuilders, this chapter focuses on the question if and how the government could support this. The UK is used as a case study, but findings may be applicable to other economic contexts.

The rationale for a role for government is assessed from the perspective of market failures (see Box 4.1).

Box 4.1. What is a market failure?

In essence, a market can be distorted by some external or structural influence which impacts its ability to function in a way where resources are allocated efficiently in consideration of broader economic, social, or environmental benefits to a society (Impact Evaluation Expert Working Group, 2012^[1]). Most markets do not match the perfect model imagined by economists, and many contain distortions which can harm the productivity, growth and resilience of economies (Criscuolo et al., 2022^[2]). In these instances, knowledgeable government intervention can be legitimised.

Government support can have widespread, complex, interrelated impacts, and previous OECD work weighing the upsides and downsides of it emphasise viewing it as ‘an ecosystem in which support flows in multiple directions and fundamentally reshapes markets’ (OECD, 2024^[3]).

In this way, government intervention can result in adverse effects. Empirically driven policymaking and robust impact evaluation measures are yet to be practiced widely at low cost (Criscuolo et al., 2022^[2]), and policies can both fail to meet objectives and result in additional negative consequences. Government failures as specific areas where government interventions are negatively impacting allocative efficiency may therefore also be relevant for assessing the rationale and best-suited strategy for government intervention.

Government intervention may be considered where market failures exist and where the benefits from government addressing these outweighs the costs, including those related to government failures.

Summary

Contextual market distortions in the commercial market.

- **Impact of supported foreign shipbuilding industries:** Significant subsidies by other select governments create a global market that is not a level playing field. More broadly, in the current evolving geopolitical situation, supply chain security has become increasingly relevant.
- **Impact of related advanced manufacturing:** The strong presence of related advanced manufacturing industries both positively and negatively impacts the shipbuilding industry.
- **Impact of defence.** Similar to related advanced manufacturing industries, defence has both positive and negative impacts on commercial shipbuilding, supporting the presence of valuable resources but contributing to unfair competition which can create barriers to those resources effectively inaccessible.

‘Public needs’ and procurement.

- Public procurement can be a way to satisfy ‘public needs’ but can nonetheless result in many costs and inefficiencies. The decision to use it as a tool is therefore a balancing act.
- There are risks of adverse market effects on both the supply and demand side: on the supply side, it may result in more stubborn domestic premiums and keep underperforming firms afloat at the expense of more productive firms; on the demand side, it may result in below market price cost of

transport becoming an expectation, making it difficult to adjust fares or otherwise sustain financing of the service in the long term. Strategic use to avoid those risks is therefore essential.

- Though there are some green-oriented procurement practices, innovation as an underlying objective of public procurement is underutilised in non-defence submarkets.

Shortage of skilled labour is a key challenge for this industry and relates to multiple market failures.

- The demand for certain skilled labour does not match its supply.
- Combined with cost-of-living trends in shipbuilding regions, this results in pricing discrepancies and has become an acute challenge which fuels a vicious cycle of declining production capabilities.
- Information and co-ordination failures in training and employment, as well as the positive externality of knowledge and experience, are key targets of intervention.
- Reputation of the industry has been connected to these challenges.

Digital transformation and future-oriented strategies are essential for ensuring competitiveness.

- Digitalisation of design and production is a key characteristic of being competitive in the global market.
- In the UK, it appears that large, defence, or high-tech start-up firms tend to be further along in digital transformation of the design and production process.
- Digital transformation requires both financial support as well as a willingness by firms to change themselves structurally and culturally.
- Digital transformation of shipbuilding and other industries is likely to change skills needs but also, potentially, overall demand for labour. Planning for this now could help mitigate future harmful production or social costs.

Innovation and emerging technology.

- Innovation is essential for long term productivity growth. Knowledge and rent spillovers justify support for innovation.
- A primary challenge in the United Kingdom has been the step from innovation to commercialisation.
- Market power and regulation as a barrier or a support are identified as influential factors.

Net Zero.

- Negative and positive externalities in relation to the Net Zero transition are likely to impact pricing of different forms of energy and broader energy security. These incoming effects are only partially accounted for within the current market, slowing offshore wind development and the transition to clean maritime shipbuilding.
- While national and nation-level Net Zero goals have been articulated, concrete steps need to be taken to reduce uncertainty in the industry and develop offshore wind and transition to clean maritime shipbuilding. The recently published Maritime Decarbonisation Strategy may be a valuable avenue.
- Infrastructure necessary for offshore wind and green vessels requires significant investment, creating a collective action problem, slowing offshore wind development and the transition to clean maritime shipbuilding.
- Additional negative environmental and social impacts from the 'green' transition are currently not significantly internalised by the industry or in the price. Where government intervention occurs, policy action which considers these externalities may carry additional benefits.

Place matters: Regional development as a strategy for supporting the shipbuilding industry may support both growth and resilience through enhancing spillover effects and market dynamism.

- Clusters function as key nodes of information and co-ordination in many shipbuilding economies, including the UK.
- Co-location of firms within a sector as well as with other related sectors is common and competitive shipbuilding economies are frequently characterised by policies targeting this cluster level. An example is the leisure sub-sector in Italy.
- The combination of multilevel governance and collaboration with industry associations can improve collaboration and performance in addressing local challenges.

Cross-sectoral knowledge and development as a strategy for supporting the shipbuilding industry may support both growth and resilience through enhancing spillover effects and market dynamism.

- Given natural interrelatedness of shipbuilding with other sectors (via a maritime economy view or in respect to the relationship with transport, energy, or advanced manufacturing sectors), a cross-sectoral perspective is rational and useful for addressing information and co-ordination failures and supporting economic resilience.

Aligning information and policy along key sectors in maritime and advanced manufacturing may promote positive spillover effects.

A. Contextual market distortions

Summary

There are three categories of external factors which impact the domestic UK shipbuilding sector and the markets it operates in. These impacts are structural and generally applicable across sub-sectors and key challenges and are relevant for considering a role for government. The three categories are supported foreign sectors and globalisation, competitive domestic advanced manufacturing sectors and the domination of defence shipbuilding.

Impact of supported foreign industries

Significant subsidies by other governments create a global market that is not a level playing field. More broadly, in the current evolving geopolitical situation, supply chain security has become increasingly relevant.

One key contextual market distortion from the perspective of a domestic economy is that of global markets and foreign industries. Even if a market is structurally set up to function efficiently and effectively within a national economy, if foreign economies impact the competitiveness of this domestic market as a whole, the domestic market is nonetheless distorted.

This is not inherently a negative consequence. For example, it is one of the significant benefits of trade among countries which allows for countries to benefit from the differing comparative advantages amongst themselves. Nonetheless, one impact of this allocative efficiency among countries trading can be that there is a redistribution of benefits and costs among the domestic producers and consumers in the markets involved or related to the trade (Rodrik, 2012^[4]). So, despite the general benefits that free trade can have, domestic redistributive effects are present.

Wage disparities. One relatively standard consideration in this context is that of wage disparities among countries. *Ceteris paribus*, relatively lower labour costs abroad impact competitiveness of domestic firms. And while global value chains have resulted to increased wages in some countries (Paweenawat, 2019^[5]; Sasidharan, 2024^[6]), they can tend to increase wage inequality among certain groups domestically (Costinot, 2012^[7]; Sasidharan, 2024^[6]; Allen, 2021^[8]; Endoh, 2021^[9]) and may also foster regional deindustrialisation among developed countries (Fothergill, 2013^[10]; Stöllinger, 2016^[11]). Recent literature on how these inter-country labour cost dynamics impact competitiveness for advanced manufacturing industries with complex supply chains – such as shipbuilding or aerospace – is relatively sparse. Furthermore, cheap labour cannot explain the significant differences in productivity between, e.g. Japan and Korea and the UK, so this effect may be a relevant avenue to explore but should not be treated as an actionable target for policy intervention.

The effects of globalisation on domestic markets can become more significant (distortive) if foreign industries are subsidised or otherwise supported in significant ways. Government support for certain industries can inflate firms' competitiveness on the global market, which can in turn impact the ability of other countries' industries to keep pace without similar support. Certain practices within this approach which lead to price discrimination, such as dumping, are regulated internationally due to the harmful effects they have on other countries' markets. On the other hand, there are many government policies which are less intentionally harmful but nonetheless impact foreign industries. Previous research on certain government support measures in shipbuilding suggest that this practice emphasise the harmful disruption to a 'level playing field' among shipbuilding and shipping economies (Gourdon, 2019^[12]).

In recent years, many countries have begun to turn to more prolific industrial policies in manufacturing, including shipbuilding (Criscuolo et al., 2022^[13]; OECD, 2025^[14]; UK Government, 2024^[15]; Lloyd's List, 2024^[16]; European Commission, 2022^[17]). This may relate to the risks associated with globalisation, such as supply chain disruptions and volatile price fluctuations, as well as broader geopolitical trends. One key characteristic of industrial policy is that markets are not only targeted for the resolution of certain distortions, but also with direction towards certain broader goals (Criscuolo et al., 2022^[2]). For many, it is also a way to drive government goals cross-sectorally, such as Net Zero or advancement in automation (OECD, 2025^[14]). In a global context of protectionist policies, sectors – if left on their own – may face certain disadvantages because foreign competitors are receiving artificially high levels of investment and support.

In shipbuilding, targeted government support for domestic industries has increased among key shipbuilding economies, including the UK. Support measures by non-Shipbuilding Committee economies have more than doubled since 2022, with the largest number of support measures provided in China (OECD, 2024^[18]). The most frequent used measures were direct transfer of funds by governments, government revenue that is foregone or not collected, and protection of the domestic market, by favouring orders by shipowners at domestic yards (OECD, 2024^[18]).

Another key measure frequently used is official export credits, whereby support is given for the purchase of ships, sometimes with prices in excess of several hundred million pounds (Box 4.2). Although export credits are provided on the contract of repayment, the amount of official export credits could be large. As the name suggests, official export credits are intended to support exports. For countries with large shipbuilding capacity and where exports are the mainstay of the shipbuilding industry, such as Korea, export credits are a useful support measure.

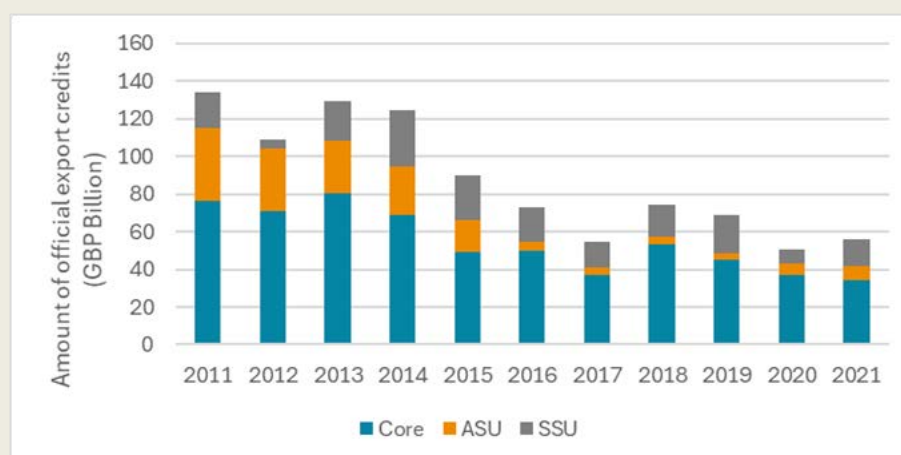
Box 4.2. Official export credits in shipbuilding

The aim of official export credits is to support national exporters competing for overseas sales through Export Credit Agencies (ECAs). ECAs can be government institutions or private companies that operate on behalf of governments.

The OECD works to develop and monitor financial discipline for export credits, as set out in the Arrangement on Officially Supported Export Credits ('the Arrangement'). Herein, the limit of how generous financial terms and conditions offered by Members may be is outlined in compliance with World Trade Organisation obligations. Several rules are sector-specific, including for ships.

Figure 4.1 shows the total amount of Arrangement official export credits based on reporting by ECAs of Arrangement Members. Official export credits under the ship-specific rule called SSU (Sector Understanding for Ships) have oscillated, averaging GBP 17 billion from 2011 to 2021.

Figure 4.1. Total amount of Arrangement official export credits



Note: "Core" business refers to official export credits supported according to the rules found in Chapter II of the Arrangement, the Climate Change Sector Understanding (Annex I), or the Nuclear Sector Understanding. "ASU" business refers to aircraft supported under the terms and conditions of the Aircraft Sector Understanding (Annex III). "SSU" business refers to ships supported under the terms and conditions of the Ship Sector Understanding (Annex IV).

Source: Export Credits, OECD, <https://www.oecd.org/en/topics/export-credits.html> [Access: March 2025], converted currency from USD to GBP.

Across Members to the Shipbuilding Committee, official export credits have been estimated to account for over 95% of all supports.

As mentioned above, selected peer competitors – as well as the UK – have also implemented support measures. A key mechanism among these countries has been through grants.

Previous work by the OECD has demonstrated the harm these types of government-driven market distortions can have on the global market and third economies (Gourdon and Steidl, 2019^[19]). In respect to the UK's current circumstances, the two primary subjects of interest are competitiveness and supply chain security.

Significant government support for foreign industries negatively impacts the relative competitiveness of the UK's domestic shipbuilders.

- Through the promotion of firms' financing and production capabilities, internal production costs are lowered and bids by foreign producers can be considerably more affordable than those at home (Gourdon, 2019^[12]).
- This is the case in global markets as well as in public procurement (discussed further in Sub-Section 4.B.).
- The cost of producing domestically therefore often includes a substantial premium, but continued production abroad may result in greater decline in production capabilities.

Both these foreign government support measures and broader geopolitical phenomena may impact current and future supply chain security.

Global supply chains have been key to productivity gains in advanced manufacturing industries such as shipbuilding, but are accompanied by risks. Supply chain disruptions over the past few years have impacted production in multiple countries, including the UK. Stakeholders have noted the impact this has had on production costs, delivery times, and filling their orderbooks. Future risks are also increasingly cited as a concern. Geopolitical tensions and trade barriers have already lengthened trade routes and increased transaction costs (OECD, 2024^[20]). The current outlook from many stakeholders is that this is unlikely to change in the near future, impacting the level of risk associated with heavy reliance on global supply chains.

Increased government involvement in industries abroad also impacts upstream aspects of production. Government support, particularly state enterprises are increasingly impacting global supply chains (OECD, 2024^[3]). Multiple stakeholders interviewed voiced concerns about key foreign suppliers being likely to switch to their own domestic priorities, adding greater uncertainty to long-term planning.

These impacts can be applied to three key materials: decarbonised steel, carbon glass fibre, and batteries.

- **Steel** – steel is a key material input and can significantly affect production costs (Gourdon, 2019^[12]). The market is generally volatile, and a majority of global revenue was generated by firms partially (25% or higher) owned by governments (OECD, 2024^[3]). UK stakeholders have previously noted supply chain security as a concern for offshore wind development (Smith, 2024^[21]).
- **Carbon and glass fibre composites** – composite materials, such as carbon- or glass- fibres or reinforced polymers, are increasingly applied in shipbuilding, especially for lightweight vessels such as powerboats, for naval ships, and for offshore structures (Rubino, 2020^[22]; Gargano, 2017^[23]). Korea has driven advancements in the development and production of these materials (Kim, 2025^[24]).
- **Batteries** – as electric vessels are gaining popularity as an option for low-emissions transport, an associated demand for batteries is observable (EST-Floattech, n.d.^[25]). Stakeholders note a key concern is the global supply chain for batteries and battery components, across which China has a significant share (European Commission, n.d.^[26]), though the International Energy Agency suggests this may somewhat diversify in the future (IEA, 2024^[27]).

Importantly, though supply chain security is a concern among stakeholders, it is an issue for which mitigation or solution-finding is largely outside the capabilities of a single firm or even group of firms. Government action may be more effective, and public efforts to improve supply chain security can ultimately benefit firms indiscriminately and without significant rivalry (OECD, 2025^[28]).

Ways in which government can support supply chain resilience include:

- anticipating and preparing for risks structurally through planning, maintaining key physical and digital infrastructures, and promoting regulatory flexibility over dysfunctional regulatory barriers;
- responsive mitigation efforts, such as domestic or internationally collaborative public procurement;
- strengthening knowledge exchange and public-private co-operation for effective co-ordination when disruptions occur;
- international co-ordination through predictable and transparent trade agreements (OECD, 2025^[28]).

From the above it becomes clear that the impacts foreign industrial policy and geopolitical tensions have on firms are ones which the government is far more well placed to address than individual firms. That is not to say that any particular government action is recommended simply out of the existence of these distorting factors, but, rather, that the responsibilities and capabilities of the government position them well, and may make it appropriate, to monitor and assess these factors, the impacts they have on the shipbuilding and other industries, and to include this knowledge in the evaluation of industry health and future policies.

Impact of related advanced manufacturing

The strong presence of globally competitive related advanced manufacturing industries both positively and negatively impacts the shipbuilding industry.

Within the domestic economy, shipbuilding does not exist in a vacuum. Other advanced manufacturing industries, such as aerospace, automotive, and nuclear energy, have a strong presence in the UK. These related industries share – compete for – certain capital and labour inputs as well a common technological basis. Because of this, the strengths and challenges in one can impact those of another. Relatedly, government intervention in one can impact the others.

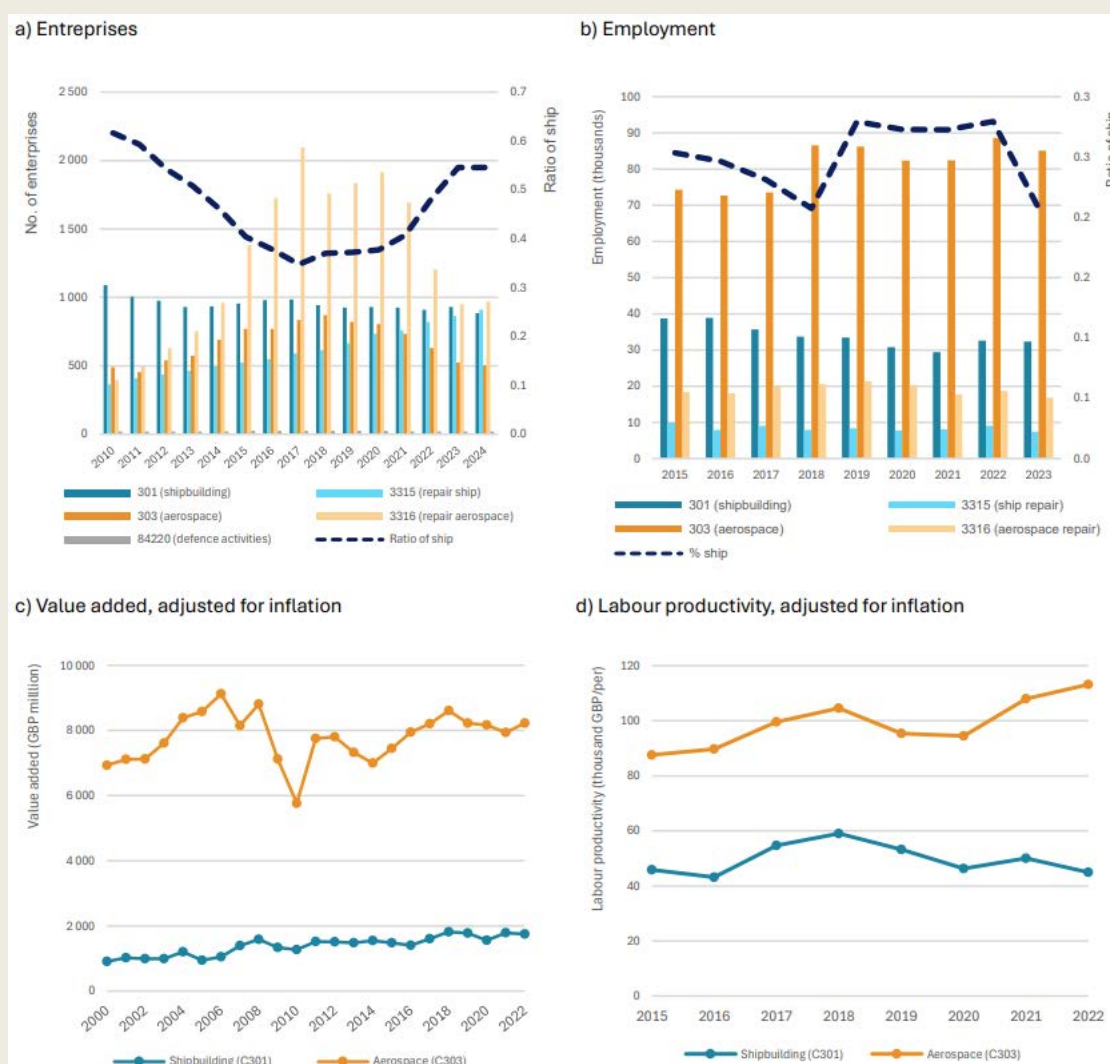
These sectors are also often co-located, which can enhance the above effects. For example, in Southwest England shipbuilding sits alongside aerospace, nuclear energy, and defence shipbuilding (discussed in the Sub-section below); in Northern Ireland, aerospace and defence dominate relative to shipbuilders.

For example, the impact of the aerospace industry has been noted by many stakeholders. Though the industries are similar in terms of many of the materials and core processes involved in production, they differ in key structural and competitive ways (Box 4.3).

Box 4.3. How does aerospace differ from shipbuilding in the UK?

Structurally, stakeholders who have worked across both industries point out that aerospace generally involves more standard orders, while shipbuilding follows more made-to-order production. This impacts organisational requirements for the production process: organising the production of multiple identical planes tends to be less complicated than co-ordinating the production of one or more customised vessels where procurement, design, and manufacturing phases may interweave.

Between 2014 and 2022, the number of aerospace enterprises outnumbered those in shipbuilding (according to ONS UKSIC-based classification of industries) (Figure 4.2 a.). Interestingly, the number of firms in aerospace building and repair has fluctuated more over time than those in shipbuilding.

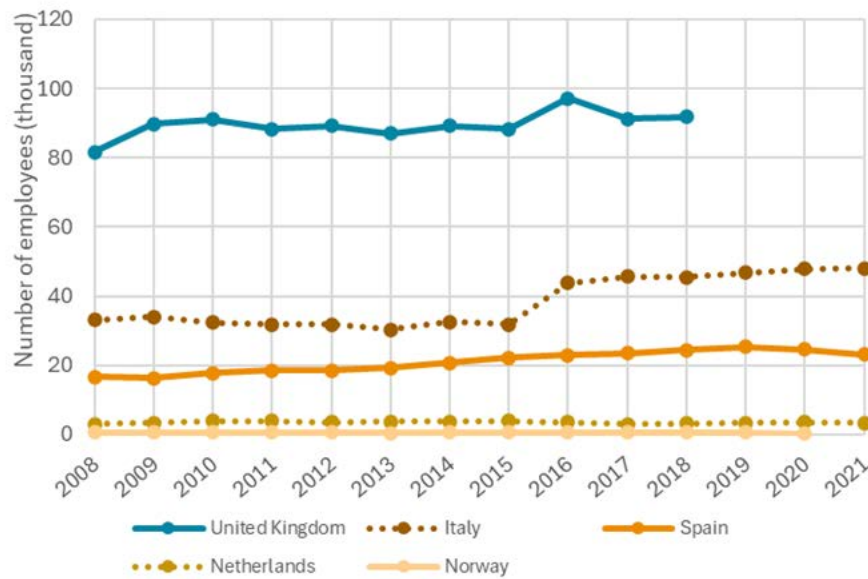
Figure 4.2. Comparison between shipbuilding and aerospace

Source: Panel a) [Nomis, Office for National Statistics \(ONS\)](#), UK Business Count [Access: March 2025]; Panel b) [Nomis\(ONS\), Business Register and Employment Survey \(BRES\)](#) [Access: March 2025]; Panel c) STAN (OECD); Panel d) OECD estimates using STAN(OECD) and BRES(ONS).

In practice, the UK's aerospace industry currently appears to be significantly more internationally competitive than its shipbuilding industry, with reportedly more consistent, long-term orderbooks and higher profit margins. This may relate to the higher value added and labour productivity reported for aerospace (Figure 4.2 c. and d.).

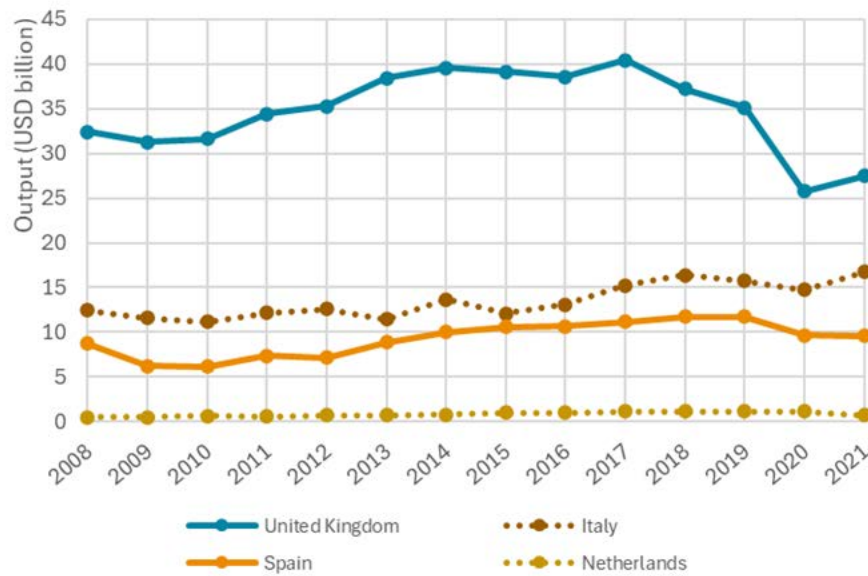
Interestingly, the UK's aerospace industry is very competitive internationally, is significantly larger than those of peer shipbuilding economies: the UK's aerospace sector boasts employment and output values that have generally been more than double than its closest competitor among the group (Italy), and the sectors are near negligible in Norway and the Netherlands (Figure 4.3 and Figure 4.4).

Figure 4.3. Number of employees in aerospace across peer shipbuilding economies (2008-2021), thousands



Source: INDSTAT Revision 4 (UNIO).

Figure 4.4. Output for the aerospace sector across peer shipbuilding economies (2008-2021), billions USD



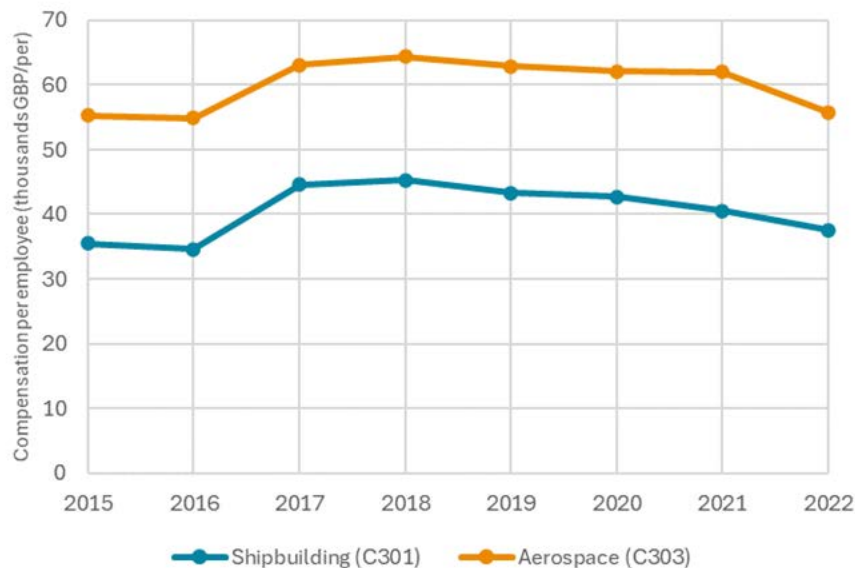
Source: INDSTAT Revision 4 (UNIO).

One aspect commonly noted among stakeholders is that these related industries can often operate at higher price points, which manifests both in relative profitability and in easing financing early stages of the production process.

Essentially, this means that on the sector-level, firms in related industries such as aerospace can afford higher input costs than shipbuilding firms, i.e. aerospace industry has greater 'buying power' for common inputs. Regarding inputs for which these industries are in indirect competition, costs can be increased beyond what they would be if shipbuilding were the only 'buyer' or downstream industry. In other words, these shared inputs are expensive for shipbuilders and impact production costs, profit margins, reinvestment opportunities, and risk assessment. However, on the other hand, competition for these shared resources could also be understood as key areas where interests align across multiple industries and where more horizontal government action may positively impact multiple (competing) sectors. This challenge and opportunity exist for both skilled labour and the broader supply chain.

Labour, notably skilled labour, is one of the clearest points of competition where the differences in price points has an impact on the relative position of shipbuilding industry. OECD estimates suggest that compensation for employees differs significantly between shipbuilding and, e.g. aerospace (Figure 4.5). The consensus of stakeholders who discussed this issue was that higher wages in, as well as the stability from a more consistent, long-term orderbook of, related industries pull labour away from shipbuilding.

Figure 4.5. Compensation per employee (2015-2022), adjusted for inflation



Note: Adjusted by CPI.

Source: OECD estimates from total compensation of employees over number of employees using STAN (OECD) and BRES (ONS) databases. Both sources do not exclude defence contracts.

On the other hand, competition also implies shared interests, especially regarding skills gaps in the current labour market (see more in Sub-section 4.C). Stakeholders in and beyond shipbuilding have noted that the limited number of workers with specific skillsets that are desired among shipbuilding and other related industries increases wages disproportionately to what firms can easily afford. Investing into programmes or other actions to increase the pool of skilled labour therefore benefits these industries, not just shipbuilding. Additionally, while competition can increase when related industries collocate in the same region, this can nonetheless pull skilled labour to the region.

Similarly, there is overlap in supply chains for shipbuilding and related industries. The differences in, essentially, 'ability to pay' among these industries impacts pricing in these supply chains which can

negatively impact production costs as well as risk assessment by shipbuilders and buyers. Simultaneously, concerns about global supply chain security are shared across these industries.

In these ways, the strong (and globally competitive) presence of related industries has both positive and negative impacts on the domestic shipbuilding industry, predominantly on the supply-side. By functioning at relatively higher price points, where these industries directly compete for inputs (e.g. skilled labour or supply chain), prices for these inputs are increased. With a global buyer's market and competitive pricing abroad (as explained in the previous Sub-section), these increased production costs can impact domestic shipbuilders' ability to secure orders and maintain a profit.

In summary:

- **The positives.** Co-existence (and often co-location) of these sectors can pool skilled labour and other supply chain resources, and encourage knowledge and rent spillovers.
- **The negatives.** That these related industries generally have greater 'buying power' to access shared resources – through global competitiveness and/or government support – increases real production costs for the shipbuilding industry disproportionately to the current profit margins firms can expect.

For government intervention, this interrelatedness is an important consideration for understanding potential intended or unintended consequences of different policy approaches. There may be greater benefit in directly addressing shared 'pinch points' than simply supporting shipbuilding through, e.g. temporary additional financing.

Impact of defence

Similar to related advanced manufacturing industries, defence has both positive and negative impacts on commercial shipbuilding, supporting the presence of valuable resources but contributing to possibly unfair competition which can create barriers to those resources, making them effectively inaccessible.

Defence also has an impact on markets relevant to the domestic shipbuilding industry. It is arguable that the designation between defence shipbuilding and commercial shipbuilding is somewhat arbitrary, especially since 'defence shipbuilding' in the UK includes both commercial exports of defence products as well as defence procurement through domestic government bodies (primarily the Ministry of Defence). However, the distinction is drawn for three key reasons: first, to cover defence aspects of an industry or country in depth is outside of the scope of the work of the OECD; second, defence markets – similar to luxury goods – are impacted in somewhat different ways than more standard commercial goods by broader political or economic changes, making the investigation of the relationship between defence markets and their non-defence commercial counterparts an interesting and relevant action; finally, relatedly, defence as a public need which has immeasurable (and debated) benefits for a society not only helps justify the previously-drawn distinction but also accounts for additional domestic supply-side considerations.

Briefly, as noted previously, data for defence is even more limited than for general shipbuilding (or other industries within which defence contracts can be accounted for). Defence as both a commercial export and publicly procured good within these industries, in terms of how bids are awarded, further complicates this as even for 'commercial' firms it can be unclear how much of their economic activity can be attributed to defence versus non-defence production. Please see the companion paper on data capabilities for more detail.

Defence shipbuilding constitutes a majority of domestic shipbuilding production in comparison to non-defence submarkets. A previous internal report for the NSO estimated the share of sales domestic shipbuilding that were related to defence to be 77.6% in 2022. This constitutes a combination of commercial defence contracts and public procurement. Furthermore, within defence shipbuilding, there

are three major players who may be said to have considerable market power. Beyond that, given current global tensions and investments into defence, shipbuilding and related firms producing for defence (domestic or export) benefit from both better price points and greater certainty of incoming orders. Firms which do both defence and other commercial work can benefit from a 'buffer' through defence which allows for a more stable future outlook.

Furthermore, in much the same way as with related advanced manufacturing, given competition for certain shared supply chains and skills labour pools, defence can generally pay more, increasing production costs and/or risks for non-defence firms. Specifically in regard to skills labour, stakeholders have emphasised higher wages, greater security of employment, and training opportunities as pulls to defence from non-defence firms.

There are also benefits to the spending (investment) capabilities of defence which many non-defence firms may not have. Even though defence firms, particularly the bigger shipyards, may have a more preferential pick of the limited pool of skilled labour, these skill gaps can still negatively impact production costs as well as planning and completing orders on time. Nonetheless, across defence and non-defence there is a shared interest in addressing these skills gaps. Training and investment into skills development by large defence firms can ultimately benefit other firms by, for example, pre-selecting apprenticeship candidates for which there is not enough room at the large defence firms for wider distribution among other defence and non-defence shipbuilding firms.

The greater spending power and security from competitiveness abroad and domestic procurement can also ease early adoption of new technologies and innovation, derisking these advancements for the broader shipbuilding industry. This is particularly relevant for digitalisation, where large and defence firms tend to be markedly ahead than many shipbuilding firms, according to stakeholder engagement.

An alternative view, taking into account the Subsection below on 'public needs', is that defence shipbuilding is used as an impetus for artificially high stimulation of the shipbuilding sector overall (Stott, 2023^[29]).

In summary, it is relevant to understand the 'pulling' effects the defence side of shipbuilding can have on commercial production, especially for access to and affordability of the supply chain and skilled labour pools. Nonetheless, the spending power and greater stability currently available to firms in defence shipbuilding can be beneficial for leading digital transformation, adoption of other new technologies, and evolving strategies for skilled labour, derisking these actions for commercial firms. Instruments which utilise this aspect may allow for interesting collaborations to catch up the broader sector.

B. 'Public needs' and procurement

A key consideration with respect to the rationale for government intervention is that of 'public needs' and societal benefit, respectively. Within the construct of market failures as rationales for government intervention, the justification is often based on some additional harm or benefit to society which is not currently accounted for in market pricing, resulting in allocative inefficiency. Industrial policy is often developed with a view to satisfy such 'public needs'. Sometimes this can mean 'catching up' a national industry to the technological capabilities of global competitors. Other times it can be in terms of addressing social inequalities or, increasingly, climate change. Recently, it has been highlighted as an effective short-term measure when significant supply chain disruptions occur (OECD, 2025^[28]).

Public or government procurement, the acquisition of goods and services by government bodies using public funds, is a targeted demand-side policy instrument. It is generally used to satisfy some 'public need' or is otherwise included in an industrial strategy for broader mission-based purposes (Criscuolo et al., 2022^[2]; Gourdon, 2019^[12]). When used to stimulate innovation, particularly in areas where demand is significantly driven by 'public needs', this tool can be used to help overcome barriers to entry and act as a 'proof of concept' before commercial uptake (Criscuolo et al., 2022^[2]; Criscuolo et al., 2022^[13]). On the

other hand, the OECD has previously pointed out the risk that public procurement poses to competition and market dynamism: through various practices, such as market access restrictions or domestic price preferences, the discriminatory preference for domestic firms can result in short-term gains in income and welfare, but can ultimately result in inflated prices for the final product and in less productive firms remaining active in the longer term (Gourdon, 2019^[12]). Generally, across manufacturing sectors, state ownership and heavy subsidisation correlates to underperformance relative to private firms (OECD, 2024^[3]). Maintenance of under-performing firms can also impact the industry's perception of fairness in procurement practices. Finally, to reach a comprehensive cost-benefit analysis via both short- and long-term views is currently still a challenge with this category of policy instruments (Criscuolo et al., 2022^[2]).

In the maritime supra-sector for the UK, public procurement functions as a devolved matter and has been used for instances where the product is itself a 'public need', such as with lifeline ferries in Scotland or domestic defence vessels, and sometimes to stimulate innovation, such as with electric ferries. Previous instances, discussed below, indicate that this approach carries both costs and benefits. Notably, the public perspective differs from that of firms.

From the public perspective, government procurement in shipbuilding has often been used where there is a 'public need' for which private financing is particularly challenging. Given the sizeable costs associated with tendering, vessels used for public transport or other public functions are frequently procured in this fashion. At least from the stakeholders engaged with by the OECD, it did not seem as if innovation has been a particularly strong incentive for a majority of publicly procured vessels (at least outside of defence). Notably, a common frustration on the public end is with the perceived high premiums charged by domestic firms. Exact information on the differences in estimates between domestic and foreign producers is not accessible for analysis, though is perhaps an area worth investigating internally.

Either way, the discrepancy between foreign and domestic prices has complicated the calculus, especially where many of the costs are up-front and quantifiable while the benefits are longer-term and difficult to quantify. Accepting these higher domestic premiums without proper qualification can result in the phenomenon described above, whereby, despite short-term firm-level benefits, domestic prices continue to rise and firms with insufficient productive capabilities are kept afloat at the expense of more productive firms. This risk was exacerbated by the fact that the previous iteration of the procurement system did not allow for the exclusion of suppliers from participating in a procurement on the basis of prior poor performance.¹

From the supplier perspective, despite the intention or theoretical benefit of procurement to lower costs and derisk, the bidding process often requires additional costs, and uncertainty has been generated through multiple instances of rejection. Across stakeholders with whom this topic was discussed, there was consensus that, compared to private contracting, the bidding process for government procurement requires more time and effort, including costly preparatory investment with the risk of no pay-off. This preparatory investment underscores the point by the OECD of potentially stimulating excess production capacity at the expense of real longer-term increases in production capabilities. More broadly within the industry, there have been multiple instances of false hope that projects would go to a domestic producer based on information related to the bid.

From interviews this could arise from discrepancies between the rhetoric of expressed policy objectives or dialogues between government and industry, and the more neutral, regulated process mandated for contract authorities. Following the publication of Maritime 2050, the National Shipbuilding Strategy, and the offshore pipeline, the expectation among the industry was that key contracts would go to domestic firms. However, these ultimately went abroad. A specific critique of the pipeline model was the published plot of individual ship orders set to come through set false hopes for certain domestic bids.

Some notable efforts have been made to improve government procurement. This includes investigations in response to significant incidents (EY, 2022^[30]), as well as legislative structural improvements.

The Procurement Act 2023 recently came into effect in February 2025 with the aim to streamline the public procurement process by allowing for fast, more efficient decisions. With its implementation, public procurement has moved towards a dynamic purchasing system, or so-called 'dynamic market'.² This 'market' essentially functions as a preselected set of suppliers, who have already completed necessary administrative processes and certain approvals. There's been an appetite to bring SMEs into these 'dynamic markets'. Previous track record can now be taken more into account in decision making, including, notably, past or current failures. Additionally, the Act has enabled easier exclusion of non-British suppliers. As the Act is only now coming into effect, there are still many unknowns in regard to how this new system will perform.

Another key aspect is the collection of more firm information, including in relation to Companies House. Though this is meant to improve objectivity and fairness in supplier evaluation and selection, the requirement to share sensitive information has been a source of concern among smaller suppliers. As such, there is a potential risk of adverse selection, at least in the short term.

In summary, there are both strengths and weaknesses to the use of public procurement in shipbuilding.

- On one hand, it is a valuable tool for supplying 'public needs', though doing so runs the risk of having adverse market effects on both the supply and demand side: on the supply side, it may result in continued domestic premiums and keep underperforming firms afloat at the expense of more productive firms; on the demand side, it may result in below market price cost of transport becoming an expectation, making it difficult to adjust fares or otherwise support financing of the service in the long term.
- While government involvement in these services can support their existence, and close collaboration among key public and private actors has been successful in addressing challenges, the use of public procurement can also increase the time and collaboration costs of repairing and replacing vessels.
- Updated legislation may help address some of the downsides of this balance.
- Furthermore, though there are some green-oriented procurement practices, innovation as an underlying objective of public procurement is underutilised in non-defence submarkets. With regard to a cost-benefit analysis that is difficult to quantify but which may be skewed towards short term costs against longer term benefits, the further integration of innovation as a core objective may contribute additional positive externalities, though perhaps at the cost of even greater procurement and production timelines.

C. Skilled labour

A broader theme within which market failures can be identified, with impacts across subspecialisations, relates to skilled labour. Market failures in skills arise when employers and/or workers underinvest in skills development, for instance because of imperfect information on the benefits of skills improvement or the risk of skills investments not paying off if workers leave a firm after training, leading to reduced productivity growth (Keep, 2006^[62]). This can provide a rationale for policy, although in some cases imperfect public educational systems can also contribute to skills challenges.

Compared with its peers, labour in the UK is expensive, which seems to relate to comparatively low labour productivity, and may be associated with a broader lag in production capabilities. Across stakeholders interviewed for this research, skilled labour shortages was one of the most consistently identified challenges in the industry, irrespective of sub-sector. Understanding and targeting related market and government failures may therefore be a key lever for widespread impact.

Unaffordable and limited skilled labour

- **Previously low investment into replacing a skilled workforce and recent limitations on foreign labour have decreased supply relative to demand for skilled labour, increasing the price.**
- **Competition from related advanced manufacturing sectors and defence shipbuilding sub-sectors with greater buying power, in the context of specific cost of living challenges, effectively create a price floor in the market for skilled labour in commercial shipbuilding.**
- **The resulting shortage can impact firms' production capabilities.**

In commercial shipbuilding, the market for skilled labour is characterised by high prices (wages) and an effective shortage.

First, because there is relatively high demand for certain skilled labour relative to its supply, which drives wages up. This is supported by stakeholder information noting both high labour costs and limited labour supply.

Certain factors have contributed to a relative decline in the supply of skilled labour. Historical lack of prioritisation in promoting skilled trades and recent restrictions on foreign labour may help explain this phenomenon.

- **Low investment.** Domestically, stakeholders note that a lack of investment in previous decades has had generational impacts on the availability and experience of British workers in shipbuilding. Generations with a more comprehensive set of desired skills as well as decades of experience are now aging out of the workforce, and even with newer training initiatives blossoming over the past few years, there are allegedly multiple generations of a gap in skilled and experienced labour between the outgoing and incoming generation. This is a challenge also noted within aerospace and defence.
- **Low access to foreign labour.** Relatedly, the UK's exit from the European Union has meant that labour for certain skills and occupations for both temporary and permanent work, has become less mobile between the UK and Europe. Heightened immigration regulations may have also enhanced barriers to labour mobility from non-EU countries. The cost of bringing in foreign labour – generally via agencies – is expensive; typically, only the large defence 'primes' can afford to do so.

Second, through cost of living and the market/buying power of advanced manufacturing and defence firms who can afford to offer higher wages, this essentially creates a price floor. Below this price (wage), skilled workers will likely either move away from the region or work for better paying jobs in related industries or defence instead. These effects are influenced significantly by location (and co-location in the case of related sectors).

- **Buying power of related advanced manufacturing and defence shipbuilding.** As already discussed, the comparative strength and productivity of related advanced manufacturing and of defence shipbuilding impact buying power in respect to supply chains, including the supply of skilled labour. This was a point emphasised very often among stakeholders across sub-sectors. Related advanced manufacturing and defence shipbuilding firms can pay higher wages, pulling skilled labour away from commercial shipbuilding. There is also competition at university level, such as between these sectors through the provision of programmes that tailor skills to the specific sector, networking events, and other engagement with universities. Applications for apprenticeships at major defence shipbuilding firms are in the hundreds, while SMEs in both the defence and commercial supply chains (which may overlap significantly) struggle to fill places. This issue may become more significant, as broader geopolitical tensions have increased investment in defence production.

- **Cost of living.** There are both economic and social costs of living in regions where shipbuilding is prevalent – beyond general challenges in cost-of-living in recent years. On the one hand, stakeholders note increased costs of housing. For example, according to Maritime UK South West, a shipbuilding cluster, the presence of Airbnbs make the cost of living higher than affordable through work at certain shipyards. On the other hand, there can be additional social or other factors associated with location which impact the desirability of the work: as some major shipbuilding facilities can be quite remote and/or located on other islands, these jobs carry additional costs associated with a commute or move. In this way a higher price may be required to compensate for the increased household costs of living and/or the negative impacts on personal utility.

From the previous discussion on production and financing capabilities, as well as the consistent message from stakeholders on this topic, it becomes reasonable to assume that there is currently a tangible limit by which shipbuilding firms can adjust wages to acquire the skilled labour they need. A gap in employment for firms impacts their production capabilities; in particular, employing labour where there is a skills mismatch impacts productivity.

The externalities of knowledge, age, and experience

- **Many experienced well-matched skilled workers are aging out of the workforce, and replacement with younger inexperienced workers carries additional costs, not least because of a lack of basic technical proficiency.**
- **Younger workers may adapt better to digital transformation, but this is contingent on a firm's investment capabilities as well as modernised educational or training resources.**
- **The made-to-order nature of shipbuilding further complicates the requirements for replacement.**

While the above challenge is already being partially addressed by investment into new skilled labour, stakeholders note the additional value of experience gained over time. That there is a (or multiple) generational gap between the incoming generation of workers benefitting from recent re-investment into skills in this sector and those very knowledgeable and experienced workers now aging out of the workforce is an issue presented by stakeholders across key shipbuilding regions, Scotland and Northern Ireland in particular. Replacing experienced workers can therefore carry additional costs beyond the effort of rehiring another experienced worker: replacement with young, inexperienced workers can require increased training and recruitment costs and may be associated with short- to longer term productivity losses if younger workers cannot be sufficiently trained to replace the outgoing generation. Utilising experienced workers to train incoming workers can be beneficial but at the cost of them working on production full-time, and these experienced workers may themselves need training in proper teaching.

Interestingly, it was also noted by stakeholders that training can take longer now than it did previously for various reasons. Some point to incoming applicants or apprentices who lack sufficient proficiency in mathematics, reading, and other subjects which should have been covered by their primary and secondary education. This is despite the fact that current general education requirements for apprenticeships are seen as too high by some stakeholders. Others identify out-dated teaching resources at colleges a contributory factor.

On the other hand, younger workers may be better adaptable to digital transformation initiatives in firms. There is generally a consensus that automation will impact the type of work in maritime sectors (Nautilus Federation, 2018^[31]). Especially where skill profiles may change because of technological advancement, that a larger portion of incoming workers is young could be an advantage, though only for firms with sufficient capabilities to both foresee what future skills they will need in respect to their plans for digital transformation and invest in innovative training programmes. This is a strategy gaining prominence in large defence firms but the rest of the industry (particularly on the commercial end) lags behind.

Finally, it should be noted that though there are many shared skills needs across the supra-sector of advanced manufacturing, there are certain aspects which are more specific to shipbuilding. Notably, the made-to-order nature of the industry – in contrast to *en masse* production of identical models, as is standard among automobile and aerospace – may require greater flexibility and creativity – an ability to engage in and enthusiasm for prototyping and navigating interrelated design and production processes. This is particularly the case for sub-sectors in emerging technologies where firms follow an organisation model closer to a tech start-up than a factory assembly line. This is an important point because it does limit the extent to which this challenge can be solved through co-ordination within the advanced manufacturing supra-sector alone.

Information asymmetry and co-ordination failures in education and employment

Three challenges arise in addressing the effective gap in skilled labour:

- First, there is insufficient information about skills and skills needs, both in terms of a harmonised skill-centred vocabulary but also in terms of data on skills supply and demand.
- Second, there are co-ordination failures among education, industry, and labour to appropriately and efficiently match agents in these markets.
- Finally, the reputation of the industry and specific firms influences motivations.

The first challenge in addressing the above-outlined issue is that there is not enough comprehensive information on labour and skills, within government or industry. Data availability on labour and challenges in measuring skills are key focus points of Reports I and II, respectively (see Chapter 2).

Improving information on skilled labour within the industry can help ‘free’ the market on skilled labour, as it becomes easier for agents (workers, firms, educational institutions, etc.) to identify gaps (opportunities). If information is clear and accessible – through, e.g. a harmonised taxonomy of skills across occupations or improved data collection and availability – the pursuit of opportunities by agents can become more dynamic and the market may be able to correct itself to some extent without additional intervention.

The second challenge is therefore a failure in co-ordination among education and industry to adequately create a new supply of labour and match it to demand across specific skills needs.

Stakeholders note a lack of programmes in institutions appropriate for developing the required skill sets. For example, there is no programme for naval architecture in Northern Ireland, something which, in a region where much of the labour is ultimately sourced more locally, likely impacts local firms where in-house design is a key component. Colleges may also feel little incentive to integrate industry input on needs.

Similarly, matching between individuals and firms for apprenticeships is challenging because of its largely unco-ordinated nature. Stakeholders point to the plethora of websites on, and complexity of application processes. While the larger firms typically tend to get hundreds of applications, SMEs struggle. This can significantly impact the supply chain within shipbuilding, which is largely composed of SMEs rather than big firms.

There are efforts to address this. One large shipbuilder has begun the practice of using its apprenticeship applicant pool to redistribute applicants not selected to its local supply chain. Many existing initiatives by industry associations centre around reducing these barriers by improving accessibility and visibility in this matching process.

These information and co-ordination failures are closely related. Without an effective shared vocabulary on skills and data that is skills- rather than occupations-based, signalling among education, industry, and labour becomes muddled. Clarity of signalling and promoting effective matching become costs carried by

individual agents, disincentivising engagement in this labour market in the presence of others where co-ordination is less costly.

Interestingly, despite programmes for aerospace being comparatively more robust in some regions where shipbuilding and aerospace are co-located, these information and co-ordination failures have also been highlighted as a challenge.

Reputation

A final challenge in respect to information and co-ordination is that of reputation.

Both in regard to certain firms and the industry overall, a negative reputation is attributed by stakeholders to be impacting employment of talent. This may relate to perceptions of the type of work in shipbuilding (which has been recognised as a challenge across countries) as well as to perceptions of the performance (or perceived under-performance) of the UK's domestic shipbuilding industry.

Conclusion

The ability to affordably employ workers whose skills are well-matched to their positions is essential for addressing reportedly low labour productivity.

Affordability is associated with the limited supply of skilled labour, cost of living and competition from related sectors and sub-sectors.

The supply of skilled labour itself has been impacted by an aging workforce of skilled workers, historically low investment into replacement, and a decrease in foreign labour availability.

Increasing this supply is complicated by a lack of co-ordination among industry and education, as well as the cost of training within firms. The former is significantly influenced by information asymmetry – a lack of information on skills and skills needs. The latter factor is influenced by the smaller number of 'veteran' employees with many years of experience, that incoming labour may lack basic educational proficiency, and the co-ordination costs of matching applicants to apprenticeships.

Where co-ordination across the supra-sector of advanced manufacturing may be relevant is in targeting the limited supply of skilled labour for shared skill profiles. However, a limitation is the additional flexibility which the made-to-order nature of shipbuilding may require of skilled workers which other advanced manufacturing industries may not.

D. Digital transformation

Industry 4.0 has impacted shipbuilding, both in terms of the design and production process as well as in terms of the technologies integrated into vessels.

The fourth industrial revolution, or Industry 4.0 (I4.0), generally encompasses the development and integration of advanced technologies from 2015 onwards into the performance of manufacturing production, including the integration of big data and networked information, information and communication technology (ICT), and digital technologies (Zhang and Chen, 2024^[32]; Ramirez-Peña, 2020^[33]). These advancements have generally led to more flexible, automated, and integrated production processes, and adoption of I4.0 in manufacturing firms has become essential to remain competitive globally.

There can be significant challenges, however, in adoption. In an assessment of AI intensity, regarding the adoption of AI, the manufacturing supra-sector ranked relatively high in terms of cost and maturity barriers and regulatory and ethical barriers, but highest for operational and skills barriers, relative to other industries (Calvino, 2024^[34]).

In shipbuilding, too, ‘digital transformation’ is occurring across design, production, and product. Digitalisation of design and production have become key characteristics of globally competitive firms. Relatedly, the autonomous vessel submarket is still nascent but can be expected to grow rapidly in the near future. The UK has a relatively strong position in innovation in this area but lags behind in terms of advanced facilities and digitalised production processes.

A recent support scheme in the UK has been **the Smart Shipping Acceleration Fund (SSAF)**, which committed GBP million under the UK SHORE programme to digitalisation of production or vessels (including autonomous vessels) (Innovate UK and Department for Transport, 2024^[35]). Projects will be in the close-out phase (March 2025) and the measures for impact evaluation were required, so it may soon be possible to review the outputs of this investment.

Digitalisation of design and production

Digitalisation of design and production is a key characteristic of being competitive in the global market.

What does digitalisation of design and production mean? In more concrete terms: the integration of various digital technologies into facilities and activities.

For example, a key development has been that of a ‘digital twin’, where the creation of a ‘live’ digital replica of the vessel allows for ongoing monitoring, design adjustments, and optimised production (Vinci-Carlavan, 2024^[36]). Digital twins can also be constructed for spaces, such as those developed for port environments in Southampton and north-west Scotland with support from the UK Hydrographic Office (Casey, 2024^[37]; Vinci-Carlavan, 2024^[36]).

A challenge with understanding the degree of digitalisation which has occurred in shipyards and shipbuilding firms is that (1) digitalisation as a spectrum on which firms lie is progressively evolving as new technologies emerge and become applicable to shipbuilding and (2) documentation of firms’ facilities and technological capabilities – both domestically and across countries – is limited. Thus, this information relies largely on other reports, proxy indicators, and stakeholder input.

A key factor of digital transformation of design and production is the significant gains in productivity. This is especially relevant for highly complex ‘engineer-to-order’ vessels (Vinci-Carlavan, 2024^[36]), but can generally apply across submarkets.

It is difficult to ascertain the degree of digital transformation in shipbuilding across countries. Stakeholder information comparing the advancement of facilities in China and Korea note their enhancement relative to those in the UK; others noted even larger UK shipyards were as much as ten years behind their foreign counterparts. However, these comments were based on individuals’ limited personal experiences abroad and should therefore be treated with caution. Comparing digital transformation in shipbuilding across countries may be a valuable point of further inquiry.

In UK shipbuilding, it appears that large, defence, or high-tech start-up firms tend to be further along in digital transformation of the design and production process (according to stakeholder engagement). One such UK firm interviewed noted how the use of ‘digital twins’ allowed for more integrated and efficient communication and problem solving between in-office engineers and on-site project management and workers.

The SSAF included the option for projects focused on digitalisation of port and shipyard operations (Innovate UK and Department for Transport, 2024^[35]).

More broadly digital transformation, particularly in advanced manufacturing, was noted as an objective in the Labour government’s Invest 2035 Industrial Strategy (UK Government, 2024^[15]).

E. Innovation and emerging technology

Driving productivity and competition, and accounting for knowledge and rent spillovers can justify support for innovation. Simultaneously, a primary challenge in the United Kingdom has been the step from innovation to commercialisation. Market power and regulation as a barrier or a support are identified as influential factors.

Innovation, the creation and dissemination of new knowledge (and technologies), is one of the most classic areas for the ‘market failure’ lens and industrial policies. As a primary driver of productivity growth, innovation as well as the commercialisation of innovation is essential for maintaining competitiveness domestically and globally, particularly in emerging markets.

Spillovers as externalities

There can be additional – spillover – benefits for an innovator’s sector, supply chain, consumer, or cluster which unfortunately do not pay off the costs of research and development on their own. Nonetheless, these broader benefits can reinforce productivity growth, resilience, and continued innovation. Governments are therefore incentivised to support innovation in key areas of interest.

Knowledge spillovers – ‘information or ideas emerging from other countries or firms that can be acquired without payment’ (OECD, 2017^[38]) – are a common (and in many ways desirable) consequence of innovation. Innovation requires significant investment by one firm, but its findings can become more widely available, whereby others can apply or adopt them without the same level of investment. As a consequence, a firm investing in innovation cannot appropriate the full benefits that result from this and hence may underinvest in such innovation (Impact Evaluation Expert Working Group, 2012^[1]). While intellectual property rights can help address this, a gap may remain, particularly in regard to the risk, time, and human capital of the research and development process, leading to lower innovation levels than desirable from a societal perspective. For example, innovation in autonomous vessels will benefit many who did not have to front the initial cost of development and may lead to insufficient incentives for firms to invest in autonomous vessel innovation.

While individual firms may be disincentivised through this effect to innovate, the innovation itself therefore has the potential to spur on additional development by other firms for whom there is now a more solid foundation to build from. This effect may also diffuse across sub-sectors or sectors. If a government wishes to stimulate innovation, it is therefore meaningful to address this by reducing disincentives of being an early or radical innovator, countering insufficient innovation from a societal perspective while at the same time allowing markets for knowledge and innovation diffusion to function smoothly.

Relatedly, while knowledge spillovers are largely in respect to other firms as innovators ‘free-riding’ on the effort that went into innovation, **rent spillovers** describe potential additional benefits to the consumer which are currently not accounted for in the price (Impact Evaluation Expert Working Group, 2012^[1]). For example, autonomous technology, once implemented, can be used in many ways beyond its original intention: autonomous vessels may significantly reduce operation costs by not only decreasing labour requirements but also allowing for more seamless co-ordination between vessels (Nautilus Federation, 2018^[31]). As more temporally distal effects, these are unlikely to be fully accounted for in the price the buyer pays. Similarly, benefits that are currently unrealised but may be in the future do not often make their way back to the developer without additional support.

Government stimuli: what is working, and what isn’t?

The OECD is a key source of analysis of innovation and technology developments, and of the role governments can play to enhance these (OECD, n.d.^[39]). Research and development can be both costly and risky, and, as shown above, innovators are unable to capture the full benefits of the effort. Supply-side

policy instruments to stimulate innovation are therefore relatively common, both to internalise these additional benefits but also to overcome the broader initial up-front costs of innovation.

Beyond shipbuilding-specific programmes, stakeholders feel that the national innovation system is not directed towards them (maritime).

There have been numerous support measures within the UK related to shipbuilding which are aligned with this rationale. For example, for innovation in green technology, a relevant source of funding is the UK Shipping Office for Reducing Emissions (UK SHORE) programme (Department for Transport. et al., 2022^[40]), which includes the Clean Maritime Demonstration Competition (CMDC) (Department for Transport and Innovate UK, 2021^[41]), the Zero Emissions Vessels Initiative (ZEV) (Department for Transport, 2023^[42]), and the Smart Shipping Acceleration Fund (Innovate UK and Department for Transport, 2024^[35]). InnovateUK's Catapult for Offshore Renewable Energy and Invest Northern Ireland have also been noted by stakeholders as providing key support.

As mentioned in Sub-section 4.D., these programmes have been associated with greater participation in innovation by firms, particularly in clean maritime technologies, and have supported the successful development and transition to commercialisation of notable start-ups.

One challenge with policy instruments like this is impact evaluation, the importance of which has been underlined by various OECD studies (OECD, n.d.^[43]). While stakeholders have noted the benefits of funding, they note that funding periods are relatively short and that challenges in commercialisation remain. Moreover, this does not inform greatly on the above-mentioned spillover effects, broader contributions to the sector or economy, and innovation strengths of the UK.

For the UK SHORE fund, Innovate UK is set up to conduct such an evaluation, with firms participating in these funds required to submit key information and metrics (Innovate UK. et al., 2023^[44]). Yet while case studies on individual projects have been available online (Innovate UK, 2020^[45]), and an interim impact evaluation was released very recently (Frontier Economics & SYSTRA Ltd., 2025^[46]). As this interim impact evaluation was published very closely before the completion of this report, full implementation of its findings was not feasible. However, the rigor of evaluation and use of evidence was positively received, and this practice is encouraged.

The Offshore Renewable Energy (ORE) catapult, on the other hand, was founded in 2013 (ScaleUp Institute, 2024^[47]). Yet while the recent 'Our Impact' report details what has been done, it includes little robust analysis in line with a comprehensive impact evaluation (CATAPULT, 2024^[48]).

For both nascent and established programmes then, there is currently little empirical information on the 'real' impact these programmes have.

Theory on innovation and stakeholder engagement suggest that these support measures likely benefit firms and promote innovation. However, without empirical evidence, it becomes difficult to assess these programmes in their current format in respect to alternative policy actions (opportunity costs). More information in this regard would help address any government failures present in the structure of current support measures and could help innovators incumbent to these programmes benefit from the lessons learned by previous participants.

The Innovation-to-Commercialisation Step: A Common Challenge

A common challenge with respect to innovation is not 'just' the innovation itself but the way this is subsequently commercialised and brought to the market.

Many firms fail to commercialise their innovation. Though there are many unknowns here, as there may be both case-specific reasons and broader systemic issues, available information suggests that production costs, uncertainty, insufficient investment incentives, and insufficient guidance for nascent firms

can act as barriers to commercialisation. The findings of the interim impact evaluation report on UK-SHORE generally align with this conclusion (Frontier Economics & SYSTRA Ltd., 2025^[46]).

Production costs of commercialisation can be significantly higher than during research and development, and hence risks may be higher. Stakeholders note longer lead times and complexities in navigating the supply chain and skills challenges as key factors. Many projects under UK-SHORE faced significant (13) or some (29) barriers in regard to procurement or supplier challenges for the delivery phase of the project (Frontier Economics & SYSTRA Ltd., 2025^[46]). Entering the commercial market also means entering into competition with defence and related advanced manufacturing sectors, for supply chains and labour, something even established firms struggle with.

Government support for innovation often focuses on R&D itself and less on commercialisation efforts thereafter. Given the high costs of moving into prototype and product development, this may hamper bringing potentially promising innovations to the market and instead leave them ‘on the shelf’. And while the benefits of the above funds have been emphasised by firms, one frustration is that the funding periods are relatively short. For example, projects for the Smart Shipping Acceleration Shipping Fund needed to start by November 2024 and end by March 2025 (Innovate UK and Department for Transport, 2024^[35]). This may contribute to the challenge in commercialisation, as there is little time and capacity to make the transition.

One characteristic highlighted in regard to those projects which do successfully commercialise is early planning to do so within the programme of government support. Proactive guidance and support beyond the financial stimulus may be important to navigating and overcoming the above barriers to entry. This included, in one case, fostering an export-orientation (in addition to domestic opportunities) early on in the programme.

It should be emphasised that there are differences among the programmes within UK-SHORE, notably based on where in the process from initial development to commercialisation a project sits, which influences the amount and length of funding (Frontier Economics & SYSTRA Ltd., 2025^[46]). Additionally, start-ups that have successfully garnered private investment, also benefitted from other funding through Innovate UK or through nations-level funding programmes. The expressed frustrations may therefore arise out of a combination of real areas for growth in these programmes as well as information and co-ordination failures.

Information asymmetries which result in co-ordination failures for gaining sufficient private (or other) investment was a noted challenge, especially by industry associations. A representative from a shipbuilding cluster noted that innovation assets are not well communicated. This is substantiated by the practical challenges experienced during the drafting of this report to find concrete, comprehensive information on innovation in the UK – new technologies, start-ups, broader successes – beyond grey literature and case studies from different government-funded programmes. A platform to match public or private investors to developers wishing to commercialise does not exist or was not found during the research process. According to an industry cluster association, firms, including SMEs, are interested in setting up a forum to support commercialisation of innovation.

Challenges in translating innovation capabilities into commercial production capabilities leads to a dilemma for policy.

As discussed in various sections above, decline in production capabilities is a challenge highlighted across stakeholders, and has largely been related to two points:

- First, next to cost factors and change in vessel type production, historically lower investment into maintenance and development of production capabilities (through, e.g. the sufficient maintenance of a domestic skilled labour pool or the introduction of more advanced facilities in shipyards) likely explain this decline;

- Second, challenges commercialising innovation play a role as well.

While the combination of public procurement and supply-side funding for innovation through a competitive environment – supporting successful start-ups over declining traditional shipbuilders – may be useful for overcoming the innovation-to-commercialisation step, this risks underestimating the relevance of the first point. Bruno and Tenold conclude that the formation of new shipbuilders as new shipyards is impossible without stimulus given the size of initial investment (Bruno, 2011^[49]). This may be a significant incentive for various actors to support underperforming shipyards, to avoid significant stranded assets.

The UK therefore faces the dilemma if policy support should focus on enhancing capacities or on production capabilities. And while there do seem to be ways to revitalise these production capabilities – primarily via digital transformation and maritime decarbonisation – they will take large investment and may still not result in comparative advantages, especially for very large vessels. Notably, those two developments may also be technological applications which create a very wide gap between those who adopt them in production and those who don't. Regarding the ability to competitively build ships, this contemporary trade-off is what led one stakeholder to indicate the industry is at 'one minute to midnight' to reverse the decline, and another to remark that, 'hope is not a strategy.'

Thus, in developing an industrial strategy for shipbuilding which is integrated into broader government directives, it is important to consider in what elements of the value chain relative strengths lie and to capitalise on that.

Regulation as a hindrance or a guide

Regulation is an important factor which may impact innovation and the commercialisation step, particularly in areas of emerging technology. **Absence of regulation, lack of clarity among existing interrelated regulations, ineffective communication between government and industry, and lagging classification for newer products or technologies can dampen enthusiasm for commercialisation of innovation, can increase production costs, and can risk unintentional noncompliance which could ultimately cause social harm.** Many projects under UK-SHORE faced significant (17) or some (25) barriers in regard to regulatory challenges for the delivery phase of the project (Frontier Economics & SYSTRA Ltd., 2025^[46]). This indicated the highest share of 'significant' barriers among the listed factors.

When regulation which is designed for, or otherwise clearly accommodating to, some nascent technology or product is absent, this can result in confusion. First, a lack of regulation can demotivate investors or operators who see a risk to safety of people or assets (Nautilus Federation, 2018^[31]). Second, in such an absence, it may be difficult to navigate which existing 'best-fit' regulations would apply, particularly if those regulations contradict one another or apply poorly to the new technology or product in practice. For example, stakeholder emphasise that some of the regulations on autonomous vessels were originally drafted for very large vessels and therefore result in requirements which are very strange when applied to smaller high-tech vessels. While this has been partially addressed by recent exemptions given by the MCA for smaller vessels (MCA, 2024^[50]; MCA, 2025^[51]), for clean maritime similar frustrations were expressed in regard to the lack of regulatory leadership.

Strengthening co-ordination among government and industry can help support both innovation and timely production. This can be on both the supply- and demand-side: on one hand the cost of production may decrease, in terms of time spent in the back-and-forth with regulators to comply with unclear requirements; on the other, on the motivations of the buyer if the regulatory landscape suggests higher risks which would be the burden of the operator to carry. In fact, in some cases, from a firm perspective, it may not be a lack but rather an abundance of regulatory pressure which may disincentivise firms to bring their technologies to the market.

Similarly, the lack of standardised categorisation of vessels or specific vessel characteristics can impact both buyers' and sellers' knowledge of the market, increasing the cost of matching buyers and sellers.

It is also worth recognising that confusing regulations can ultimately result in unintentional noncompliance, which could ultimately cause harm to workers operating vessels, passengers, or other product ‘users’.

F. Net Zero

Another key objective for which many countries are utilising industrial policy is that of sustainability and ‘Net Zero’, meaning the transition to low- or zero-carbon emitting economies. Positive and negative externalities in this area are important considerations for a role for government as well as related required infrastructural shifts.

Emerging multinational Net Zero policies, not internalised

International and national Net Zero policies for shipping are likely to affect operation costs, as well as vessel type demand, in the near future. Relatedly, the demand for offshore wind energy is likely to increase because of various Net Zero objectives domestically and abroad.

International Net Zero policies are relevant for possible further industrial policies towards shipbuilding in various ways.

- First, they affect the types of vessels that need to be built, retrofitted and operated, with the question being to what extent industrial policy is required to enable this transition.
- Second, they help to create new markets, such as for offshore wind, which in turn provide opportunities for UK shipbuilders and a possible role for the government to accommodate that.
- Third, where such Net Zero policies include the pricing of emissions they may increase costs for shipbuilders, a challenge in this highly competitive market.
- Finally, Net Zero policies come with significant investment needs for infrastructure – necessary for, e.g. offshore wind and green vessels – and a possible role for government.

The EU’s new regulations on shipping emissions provide an example (Box 4.4).

Box 4.4. New EU Regulations on Shipping Emissions

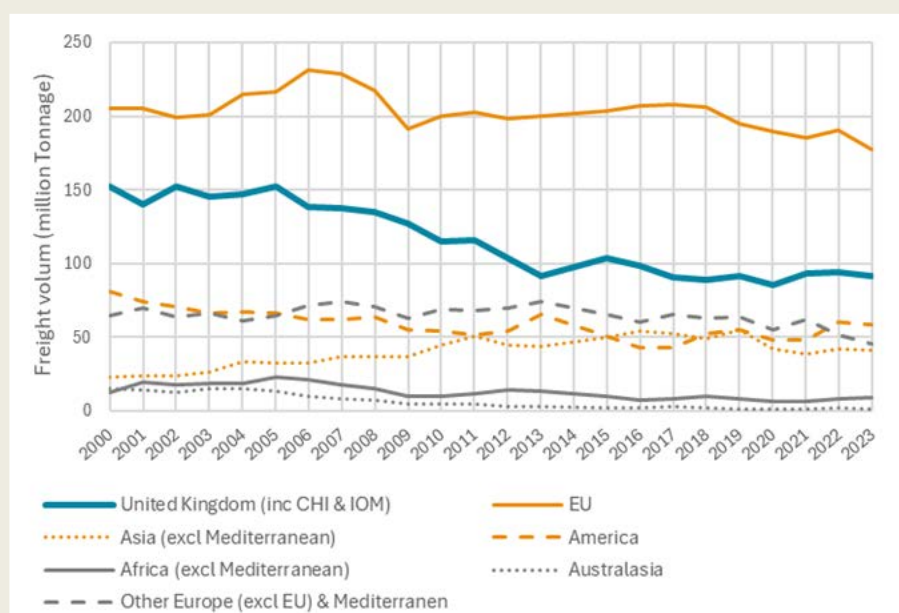
As the UK has a close economic partnership and economic interaction with EU Member States, the country is also affected by the FuelEU Maritime Regulation recently adopted and fully applied since January 2025 (European Union, 2023^[52]). FuelEU Maritime establishes maximum limits for the annual average GHG intensity of energy used by ships over 5 000 gross tonnage calling at EU ports, aiming for a gradual reduction in GHG emissions. Targets start with a 2% reduction by 2025, increasing to 80% by 2050, and cover different GHG: CO₂, methane, and NO_x emissions on a Well-to-Wake (WtW) basis.

Since January 2024, shipping emissions have also been included under the European Emission Trading System (EU ETS) (European Commission, 2024). In effect, all ships over 5 000 GT trading within the EU will have to purchase and surrender greenhouse gas (GHG) emissions from their travels. The share of emissions covered by allowances gradually increases, from 40% in 2025 of emissions reported in 2024, to 70% in 2026 of emissions reported in 2025, up to 100% in 2027 and beyond of reported emissions. Ships included under the EU ETS will have by 2027 to purchase and surrender 100% of reported emission for within the European Economic Area (EEA) routes, and 50% between EEA and non-EEA ports.

Figure 4.6 shows the main freight routes of the UK. Those with the EU has consistently constituted the largest freight routes for inward and outward exchanges with UK ports, totalling 41.8% of freight handled

in UK ports in 2023, far beyond any other trading partners. Hence, given the extensive exchanges between the UK and EU member states, this levy on 50% of emissions of UK vessels voyages is likely to put increasing pressure on UK operators to decarbonise their fleet. Failure of shipping companies to decarbonise will increase the operational expenditure of those firms in the long term and decrease their competitiveness.

Figure 4.6. Volume exchange from the UK by region (2000-2023), millions tonnage



Source: Maritime statistics: interactive dashboard, Department for Transport (DfT), Government of UK, <https://maps.dft.gov.uk/maritime-statistics/index.html>.

However, while low- or zero-emissions capable vessels might be a worthy investment for companies in the future as regulations might become more stringent, current investments into these technologies entail a large cost premium that might not be fully internalised in the short term.

Given these existing risks, financial actors might be unwilling to engage in these risky loans, limiting access to capital (or with very high cost reflected by the interest rate) for shipping actors. In the context of this large costs premium and uncertainty tied to investment in the green transition, the state can act as a de-risking actor, by providing subsidised loans or guarantee, notably under the existing term of UK innovate loans, or through Export Credit guarantees.

Clear policy vision and implementation

One trend in 'green' industrial strategies is that while the ultimate goal is often clearly articulated, the step-by-step route to achieve objectives – such as zero emission shipping by 2050 – is left unclear, particularly in terms of the development of the technological, economic, and social capacities to realise those objectives (Criscuolo et al., 2022^[2]).

The UK has generally shown strong leadership in legally binding net zero targets and various strategic plans (IEA, 2024^[53]). However, a majority of these have not engaged deeply with shipping and shipbuilding.

Following updated targets by the International Maritime Organisation (IMO), the UK has committed to zero-emissions shipping by 2050, with additional interim targets for 2025 and 2035.

Box 4.5. International Maritime Organisation Measures

In 2023, the IMO revised its GHG strategy to “reach net-zero GHG emissions by or around, i.e. close to, 2050”, along with several checkpoints, of at least 20% (striving for 30%) by 2030 and 70% (striving for 80%) by 2040, compared to a 2008 baseline (IMO, 2023^[54]). As mentioned previously, it has also set targets for uptake of zero or near-zero GHG emissions technologies fuels and/or energy sources to make up at least 5% - striving for 10% - of the energy used by international shipping by 2030.

In addition to this revised strategy from 2023, IMO members are in dialogue regarding possible mid-term measures, now called Net Zero Framework, under a proposed Chapter 5 of MARPOL Annex VI, consisting, possibly, of a goal-based maritime fuel standard, and a maritime GHG emissions pricing mechanism. The latter – currently debated – could potentially include a carbon levy that would provide significant incentivisation to a GHG emission reduction as well as an increase in shipping costs

UK shipping operators could therefore soon experience additional incentives that will push demand for emission saving technologies and transition toward low- or zero-emission capable vessels.

The UK has progressively moved through different commitments in recent years. In 2019, Maritime 2050 included a focus on clean maritime (Department for Transport, 2019^[55]). The National Shipbuilding Office set the ambitious target of all new vessels for UK waters being designed with zero emissions capabilities by 2025, and widely available zero-emission fuels across the UK by 2035 (National Shipbuilding Office, 2022^[56]). In 2023, the Offshore Wind Net Zero Investment Roadmap was published: though it may have provided guidance for offshore wind itself, offshore vessels received very little attention (UK Government, 2023^[57]).

The new UK Maritime Decarbonisation Strategy, launched in March 2025, provides further direction and support for this transition (Department for Transport, 2025^[58]). This new strategy was published mere days before the submission of this report, so in-depth analysis was not feasible. Generally, it seems to serve as a useful instrument providing policy outlook, with commitment to different regulatory measures across key policy areas and aligns with the issues identified herein. However, it also admits to many remaining uncertainties in the exact measures to be implemented (Department for Transport, 2025^[58]). From the feedback given by stakeholders, the key may therefore be that the commitments made in the strategy are ultimately backed by action, creating trust in planning and guidance.

However, it should be noted that the policy focus with respect to maritime decarbonisation has primarily been on maritime transport and (to some extent) infrastructure, with relatively less emphasis on policies that support green innovation by shipbuilders themselves. Since the realisation of maritime emission reduction objectives can only succeed if the vessels are produced that make this possible, there may be a further role for industrial policy towards the shipbuilding sector to accommodate this. Notably, the planned utilisation of the NSO for government procurement related to maritime decarbonisation in new Maritime Decarbonisation Strategy may be beneficial for the reasons expressed in this and the above sections (Department for Transport, 2025^[58]), particularly if there is an emphasis on ‘proof of concept’ for innovative vessels.

The general message from the stakeholders interviewed is that in terms of policy in action, the policy outlook for renewable energy and clean maritime is not clear enough and without effective power. There were hopes, for example, that the vision laid out by Maritime 2050, popular among stakeholders, would receive legislative backing, which it didn’t.

This has stalled development in both areas. For example, stakeholders noted that the lack of clarity on which route maritime decarbonisation would take – in terms of electric and choice of alternative fuel – was

a source of caution in the industry. Some countries, like Korea, have provided guidance on which route should receive priority (ammonia in the case of Korea (Jin, 2025^[59])).

This is contrast to, for example, the UK's automobile industry, where the Zero Emission Vehicle (ZEV) legally requires producers to sell progressively greater ratios of ZEVs between now and 2035, and pay the price if they fail to meet targets (EVA England, n.d.^[60]). Something this stringent has not been implemented for vessels but may be a useful policy tool to observe in effect for inspiration.

Nonetheless, it is likely that as EU regulations go into effect and the IMO formalises its mid-term measures, impacts will be felt through shipping operators. However, because of the progressive nature of this, these pinch points will not be felt now. Still, shipbuilding – and innovation – takes time. Policy actions which drive for an earlier transition (with more gusto than the market currently is) could help minimise future costs to operators and support the industry strengthen its innovation and production ahead.

The maritime industry is undergoing very fast innovation cycles, but high uncertainty over the right path to take for decarbonisation remains. Shipowners and operators make investment decision based on the information available to them. Yet, information on upcoming regulations, decisions of other actors might not be readily available. This imperfect information at the time of decision making can create stranded assets, as uncertainty about the upcoming environmental regulations and state support could change the path of the transition, thereby putting at risks prior investments.

Infrastructure development

The infrastructure necessary for offshore wind and green vessels requires significant investment, creating a collective action problem, slowing offshore wind development and the transition to clean maritime shipbuilding.

Both for offshore wind (Box 4.7) and green vessels – electric (Box 4.8) and alternative fuel (Box 4.8) –, large-scale infrastructure development is necessary.

Box 4.6. Infrastructure requirements for offshore wind

One major requirement for offshore wind is expansion of the power grid. The International Energy Agency has noted that expansion and integration of grid infrastructure is an area of 'critical priority' (IEA, 2024^[53]).

According to the Transmission Acceleration Action Plan (Department for Energy Security and Net Zero, 2024^[61]), the demand for electricity in the UK is expected to be a doubling driven by the expansion of electrification and decarbonisation for heat, transport and industry in the transition to net zero.

In order to meet these demands, it is essential to not only increase power generation capacity, but also to strengthen the grid infrastructure, i.e. the facilities for transmitting the generated power to users. The Plan acknowledges that the UK requires around four times as much new transmission network in the next seven years as we have built since 1990.

The electricity generated by offshore wind turbines is generally transmitted from the offshore wind turbines via cables to substations, and then to the existing power grid. According to the estimates of Holistic Network Design (National Grid ESO, 2020^[62]), the costs of the offshore network infrastructure required in the recommended design would be approximately GBP 32 billion. Furthermore, it is also highlighted that GBP 21.7 billion is required to be delivered by 2030 for the onshore power grid.

Box 4.7. Infrastructure requirements for electric vessels

For electric vessels onshore power supply infrastructures are required, including the use of electricity grid to replace the reliance of ships on their auxiliary engine when at ports (Raucci, 2019^[63]).

Currently, out of the 160 ports referenced in Clarksons port data base, only 6 had active onshore power supply, Aberdeen, Belfast, Portsmouth, Southampton, Montrose and Leith. In 2022, as part of its maritime decarbonisation strategy, the UK DfT launched a call for evidence on onshore power. There was a wide support among maritime stakeholders for the uptake of this technology to reduce at berth emissions, with nearly $\frac{3}{4}$ of respondents being supportive (Department for Transport, 2023^[64]).

Box 4.8. Infrastructure requirements for alternative fuel vessels

Despite having the second largest ports industry in Europe, only 7 (6.2%) of UK ports identified in Clarksons have potential or active alternative fuel bunkering infrastructures (Clarkson Research, 2024^[65]). Comparatively, 14 Spanish ports, including all 10 largest in terms of ports calls number in 2023 had active LNG bunkering infrastructures; all 8 largest Ports in the Netherlands had active LNG bunkering and other potential alternative fuel bunkering infrastructures, including ammonia, methanol and hydrogen. As a whole, countries in the EU had in 2023 84 ports with active LNG bunkering infrastructures, and 3 with active methanol bunkering infrastructures.

Only considering active bunkering infrastructures, less than 5% of UK ports can currently accommodate vessels running on LNG, and none of them have active infrastructures to bunker other types of alternative fuel.

Looking more specifically at the distribution of alternative fuel bunkering readiness across port size in terms of calls in 2024, only one of the 5 largest UK ports in terms of port calls number in 2024, namely Immingham, had active alternative fuel bunkering.

Table 4.1. Largest UK ports by port calls in 2024, and alternative fuel bunkering infrastructures

Port Name	Port Calls (in number)	Alternative Fuel Bunkering Active	Alternative Fuel Bunkering Potential
Dover	13 832		
Aberdeen	5 945		Hydrogen
Belfast	5 518		
Immingham	5 003	LNG	
Tilbury	3 436		

Source: OECD calculations based on Clarksons Research Services Limited (February 2025).

As a study by the Global Maritime Forum along with UCL Energy centre, highlighted 90% of the required investments globally will be onshore, 44% for the alternative fuel production capacity, and 43% for the synthesis storage and distribution (Krantz, 2020^[66]). The issue of storage and distribution of alternative fuel is particularly important as certain types of fuel, notably ammonia and hydrogen require specific safety requirement and infrastructure. In a study published in 2020, the IMO attempted to quantify for a medium pilot bunkering fuel infrastructure project launched in 2019 the yearly additional costs of bunkering these alternative fuels. For methanol, the additional costs amounted to EUR 522 000 per

year, while for ammonia the additional costs went up to EUR 2 980 000. These additional costs arise both from the need of high capital expenditure for infrastructures and also higher operational expenditure than conventional fuel as it requires additional safety procedures for handling and training for workers (IMO, 2020^[67]).

In the absence of these upstream investments in infrastructures, the downstream investments by shipping owners or operators will be limited, as these decisions need to be based on long-term viable business case. Hence, investments in infrastructures are key in providing such long-term view to investors to decarbonise the fleet.

However, the organisation and required funding for this infrastructure is significant. While infrastructure development for ports is already a massive undertaking, grid development requires even more planning and co-ordination across many different agents. In the case of investments in the green transition of shipping, the European Environmental Agency refers to a “chicken and egg problem” whereby shipping actors’ decision to invest in low- or zero-emission technologies depends on the development of supporting infrastructures by other actors, especially ports (European Environment Agency, 2017^[68]). Ports, on the other hand, might not want to invest in these infrastructures until they receive a credible signal from the demand side.

This collective action problem as a barrier to maritime decarbonisation has been previously been identified in the Department for Transportation to justify public intervention in port and grid infrastructure developments (Department for Transport, 2019^[69]; Department for Transport, 2019^[70]; Department for Transport, 2025^[58]).

Negative environmental externalities, not internalised

Historically, negative impacts by production or operation of vessels on the environment or social conditions have not been internalised in the price of the good. Net Zero strategies have contributed to some of these negative externalities being accounted for. Nonetheless, it is important not to oversimplify in how comprehensively negative environmental externalities can be addressed by to clean energy or the green transition.

Furthermore, there are potential negative environmental or social impacts from green vessels or offshore wind can themselves. For example, there is increasing evidence across the globe that offshore wind development impacts ecosystem biodiversity in negative ways, particularly during construction (Watson, 2024^[71]). Plans for offshore wind farms in the North Sea may infringe on existing marine protected areas (Püts, 2023^[72]). Similarly, case studies on the local impact of offshore development emphasise the role of community acceptance in minimising risk and promoting timely project completion (International Finance Corporation, 2024^[73]).

From stakeholder engagement, it is clear that these additional environmental and social externalities are not priority factors: energy costs, port infrastructure, and planning are more tangible problems for firms and industry associations. Because of this, it is likely that accounting for these externalities may rely on government more heavily than other factors.

G. Places

Regional development as a strategy for supporting the shipbuilding industry may support both growth and resilience through enhancing spillover effects and market dynamism.

Co-location of firms within a sector as well as with other related sectors is common and often beneficial. Marshall identified three forces of agglomeration: pooling labour, sharing a supply chain, and knowledge

spillovers (Marshall, 1890^[74]). Since then, other factors have been identified, including how clusters can enhance resilience and promote beneficial competition (Faggio, 2017^[75]; Jacobs, 1969^[76]; Porter, 1990^[77]).

Clusters function as key nodes of information and co-ordination in many shipbuilding economies, including the UK (see also Report I). This involves working closely with local and regional authorities and fostering stronger networks within the industry. Much of the stakeholder information in this report has relied on input from these cluster organisations, who have a better and comprehensive bird's eye view of their region than could otherwise be gained through select interviews with individual firms over the span of only two months.

Clusters often implement their own initiatives for addressing key challenges, such as innovation and skills. While this is occasionally in collaboration with government, many of these programmes are stand-alone and self-sustaining work by the cluster and benefit from existing closer ties with the local community.

Competitive shipbuilding economies are frequently characterised by policies targeting this cluster level. An example is the leisure sub-sector in Italy. In the UK, support from the Department for Transport has previously been used to support the formalisation of maritime clusters under Maritime UK and their integration with local communities and businesses (Maritime UK, n.d.^[78]; Department for Transport, 2024^[79]).

More information is needed on cluster-level policies and their impact. Faggio et al. note the heterogeneous nature of agglomeration across different industries – different factors exist at different levels – and caution against the extrapolation of cases in one industry or context to another (Faggio, 2017^[75]). A comprehensive impact evaluation of the various instances of funding for UK maritime clusters could not be found during the drafting of this report, though the government has conducted impact evaluations for, e.g. the development of cyber security clusters (UK Government, 2023^[80]).

The combination of multilevel governance and collaboration with industry associations can improve collaboration and performance in addressing local challenges.

H. Cross-sectoral Approach

Cross-sectoral knowledge and development as a strategy for supporting the shipbuilding industry may support both growth and resilience through enhancing spillover effects and market dynamism. Given natural interrelatedness of shipbuilding with other sectors (maritime economy view, transport and energy relations due to relevance of ferries and offshore), a cross-sectoral perspective is rational and useful for addressing information and co-ordination failures and supporting economic resilience.

On both the demand- and supply-side, shipbuilding is interrelated with various other sectors, often through a combination of geographical, economic, or technological relationships. Sometimes it may be as simple as the relationship between offshore wind energy, offshore wind floating platforms, and offshore wind vessels: demand in the first stimulates supply for the second and the third. Along that chain, commitments earlier in the chain can allow for planning downstream. In other instances, it may be more complicated, such as the interrelatedness described in Section 4.A. among commercial shipbuilding, aerospace, and defence. Nonetheless, that shipbuilding interrelates to so many valuable domestic sectors is a point which complicates an easy, objective value-based assessment.

Aligning information and policy along key sectors in maritime and advanced manufacturing may promote positive spillover effects.

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Notes

¹ The Procurement Act 2023 has allowed for greater consideration of previous performance.

² As a system of doing things, independent of the economic term of market dynamism.

Peer Review of the United Kingdom Shipbuilding Industry

This report, developed in collaboration with the United Kingdom's National Shipbuilding Office, provides an in-depth assessment of the opportunities and challenges facing the UK shipbuilding and ship repair industry. Building on detailed analysis of global market trends, peer economies and domestic capabilities, the report identifies priority segments where the UK could strengthen its competitive position and enhance long-term resilience. The study underscores the importance of improving data availability, skills intelligence, and knowledge exchange to support effective decision-making across government and industry. It also examines the role of government in addressing market distortions, supporting related industries, fostering innovation, and advancing the digitalisation and decarbonisation of the UK maritime sector. The report offers a set of evidence-based recommendations to help revitalise production capabilities, reinforce the skills base, modernise facilities, strengthen supply chains, and create a predictable environment that enables strategic investment and sustainable growth.



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